API Reference | PUBLIC
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Service Ticket Intelligence API Reference

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1 About This Document

Explore the Service Ticket Intelligence API.

Learn about the second generation of the Service Ticket Intelligence API using Service Broker and XSUAA Service in SAP Business Technology Platform. The target group is composed of developers of a customer relationship management (CRM) system who need to integrate automated ticket classification capabilities into their system.

Related Information

API Version [page 5]
API Previous Versions [page 5]
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API URL Structure [page 19]
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Glossary [page 175]

1.1 API Version

The current API version is: 2.0.0. (Service Ticket Intelligence 2.0 Major Release: 1807)

1.2 API Previous Versions

See below the list of the API previous versions:

- Version 0.0.7 (Service Ticket Intelligence 2.0 Continuous Improvement: 1806 beta)
- Version 0.0.4 (Service Ticket Intelligence 2.0 Beta Release: 1804 beta)

2 Term Definition

Related Information

Confusion Matrix Object [page 6]
Model Status [page 7]
Correlation ID [page 9]

2.1 Confusion Matrix Object

A confusion matrix object shows the counts of classifications made by the machine learning algorithm versus their true classifications in a square (n*n) matrix of integers. In turn this is used to calculate the accuracy of the model, together with many other metrics that are used to improve the model. In serialized form, Confusion Matrix is expressed as an array pair containing the labels as well as the matrix itself.

The Confusion Matrix object name in JSON is `confusion_matrix`.

Attribute	Type	Description
`labels`	Array<String>	Arrays of tuples containing the classification labels. **i Note** The order is important in this array.
`values`	Array<Array<Integer>>	The confusion matrix values. The matrix rows (first dimension) represent the real labels and the columns (second dimension) represent the predicted labels.

Example

The following is an example of a confusion matrix JSON fragment:

```
confusion_matrix : {
```

```
    labels: [
        'cat',
        'dog',
        'rabbit'
    ],
    values: [
        [5, 3, 0],
        [2, 3, 1],
        [0, 2, 11]
    ]
}
```

In this example, there were eight actual cats in the data set (`5` + `3` + `0`). Out of these, five were correctly predicted as cats but three were classified as dogs. Similarly, there were six dogs in the data set (`2` + `3` + `1`), but two were classified as cats and one was classified as a rabbit, so three were correctly classified as dogs.

2.2 Model Status

See descriptions of the different statuses of the machine learning training model.

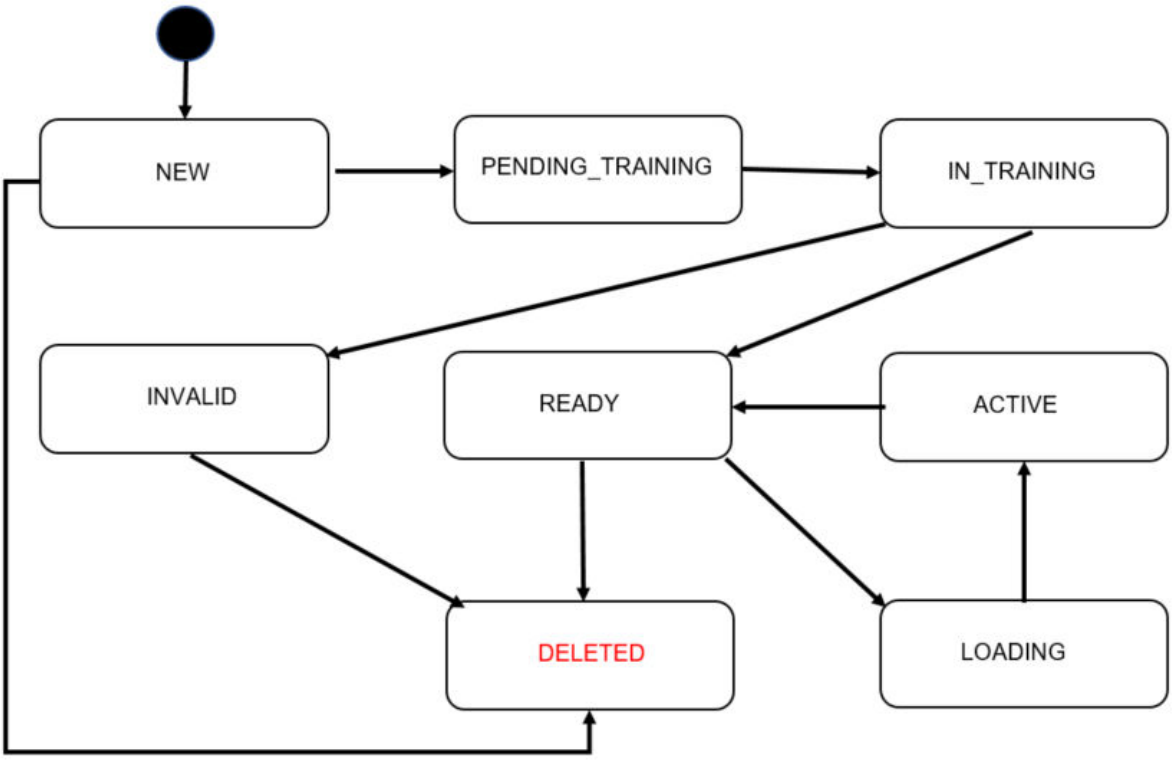
Status	Description
ACTIVE	The model is currently being used to make classifications. There can only be one model in ACTIVE status for a given business object and language pair.
DATA_EXTRACTION_FAILED	The model created by the data extractor request is invalid because the data extraction process failed.
DATA_EXTRACTION_IN_PROGRESS	The model was created but the data is being extracted.
DATA_EXTRACTION_PENDING	The model is queued for data extractor.
DATA_PREPROCESSING_IN_PROGRESS	The data extracted from OData is being preprocessed.
DATA_PREPROCESSING_PENDING	The model is queued for data preprocessing after data extraction is completed.
DATA_UPLOAD_IN_PROGRESS	The data is being currently uploaded into the model.
DELETED	The model has been removed and it is no longer available for activation.
IN_TRAINING	The model is currently being trained.
INVALID	The corresponding training file to construct the model could not be validated, or the training process was cancelled mid-way.

Status	Description
LOADING	This is an intermediate state in which the model is being activated (currently being loaded). The loading process requires approximately 180 seconds.
NEW	The model was created but training has not started yet.
PENDING_TRAINING	The model is queued for training.
READY	The model is fully constructed and can make classifications. It is ready to be used.

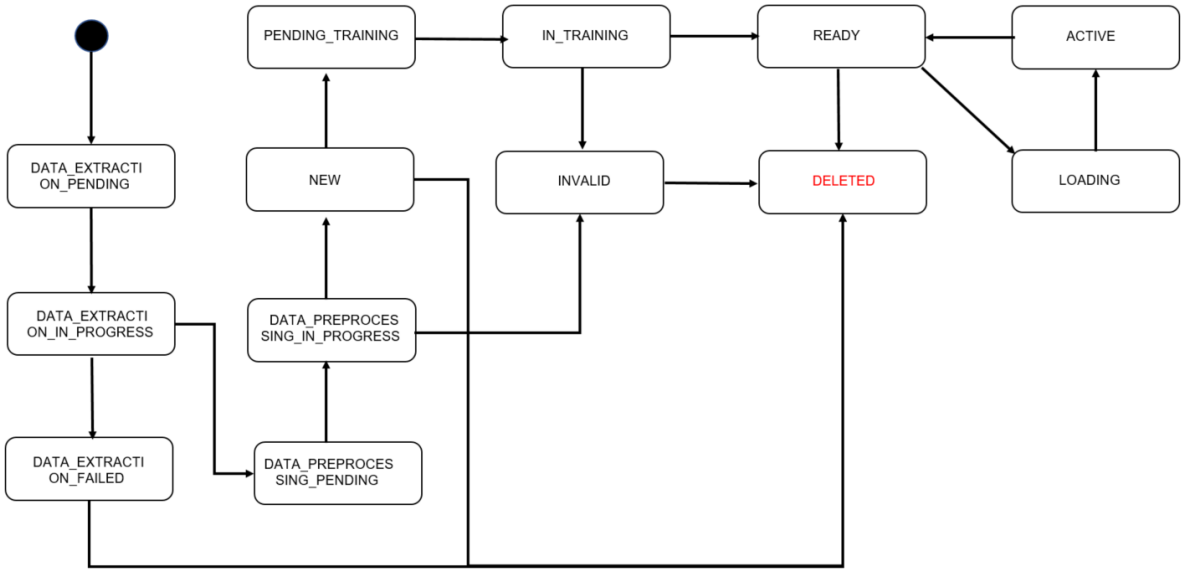
Related Information

Model Status Transition Diagram for File Upload [page 8]
Model Status Transition Diagram for OData Extraction [page 9]

2.2.1 Model Status Transition Diagram for File Upload



2.2.2 Model Status Transition Diagram for OData Extraction



2.3 Correlation ID

A correlation identifier uniquely distinguishes each request to the microservice. The client should generate this value and keep it as part of its application logs. In turn the server also logs the same value. The two logs can then be correlated, enabling end-to-end tracing.

Every request to Service Ticket Intelligence should contain a correlation ID value in the X-Request-ID HTTP header. This is a 64-byte maximum length alphanumeric field generated by the client and logged by the microservice as part of normal request logging. Clients should ensure that the value is unique for every request and is logged at their end. If the request originates from user-interaction action. For example when an administrator performs a set active model operation (see Activate Model [page 85]), the correlation identifier should also enable eventual identification of the user as well as determining when the operation was performed.

Implementing correlation IDs correctly enables accountability logging. Both read and write access to the microservice becomes traceable to the individual operator without needing to be aware of the user databases of the client systems.

Example

The sample HTTP request below shows the client activating a model, and provides a value in the X-Request-ID HTTP header.

```
PUT /sti/training/model/activate HTTP/1.1
```

```
Content-Type: application/json
Authorization: Bearer 9E00D746A498421D82C379A54BE4565C
X-Request-ID: CD5DB333D8E24C86AA6EB0D75EBF35EF-2-1
{
    "model_id": "32E78EFCB8184240838C120F0197640C"
}
```

In this example, the correlation ID value is the client's business transaction identifier, followed by the step number, and then followed by the request number. This structure is not enforced by the microservice and would be logged as it is (the value is logged without further processing or checking).

3 Recommended Workflows

The following sections describe the recommended approach to use the API calls presented in this document. However, client systems are free to implement other approaches according to their needs.

Related Information

Model Training [page 11]
Model Activation [page 12]
Classification / Recommendation Inference [page 13]

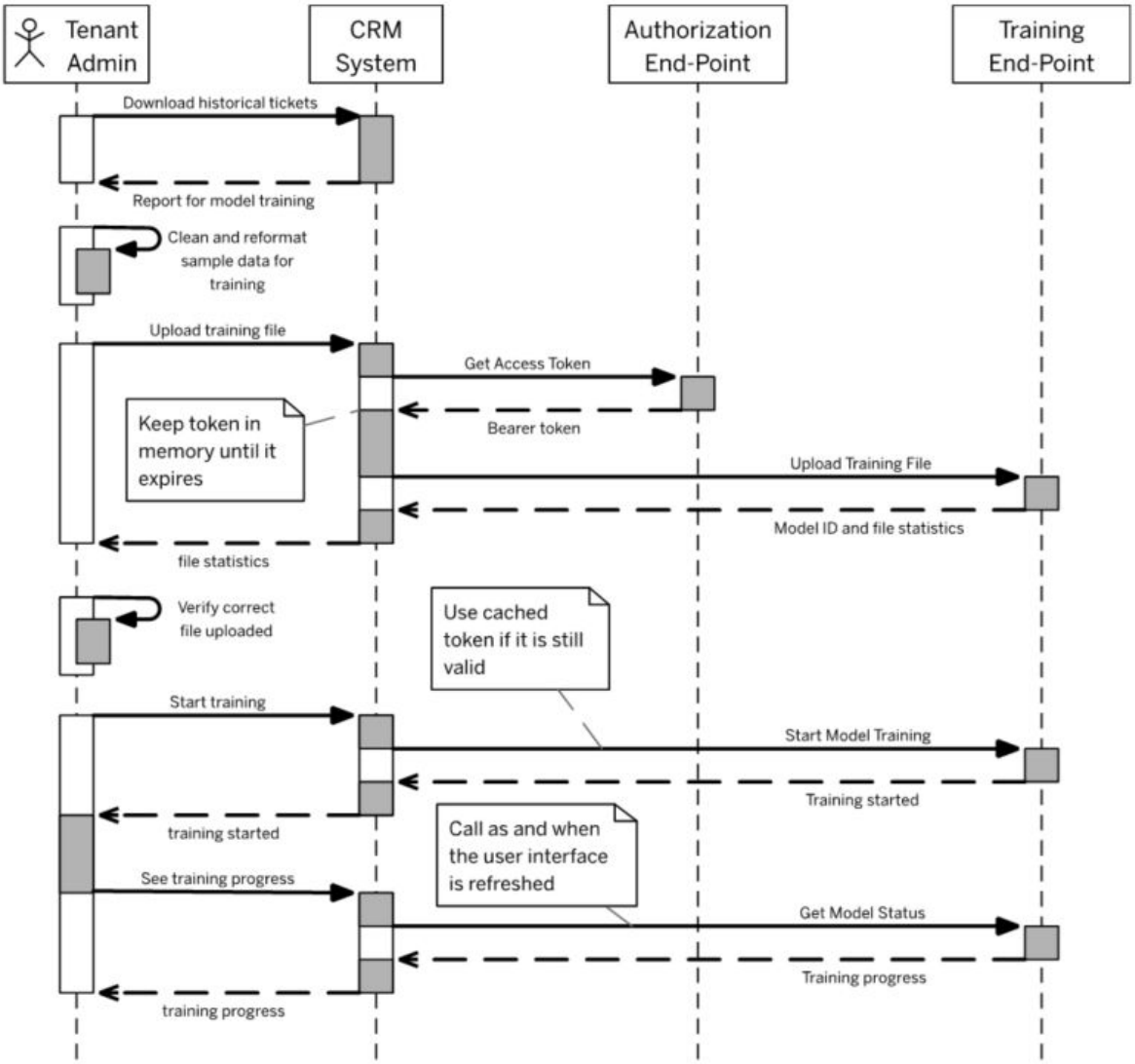
3.1 Model Training

The model training workflow takes in historical tickets and their respective classifications and creates a machine learning model from the data. In turn, the model is used to automatically classify subsequent tickets coming into the system.

Related Information

Diagram for Model Training [page 12]
Get OAuth Token [page 17]
Upload Training File / Data Extraction [page 20]
Get Model Status [page 66]
Start Model Training [page 83]

3.1.1 Diagram for Model Training



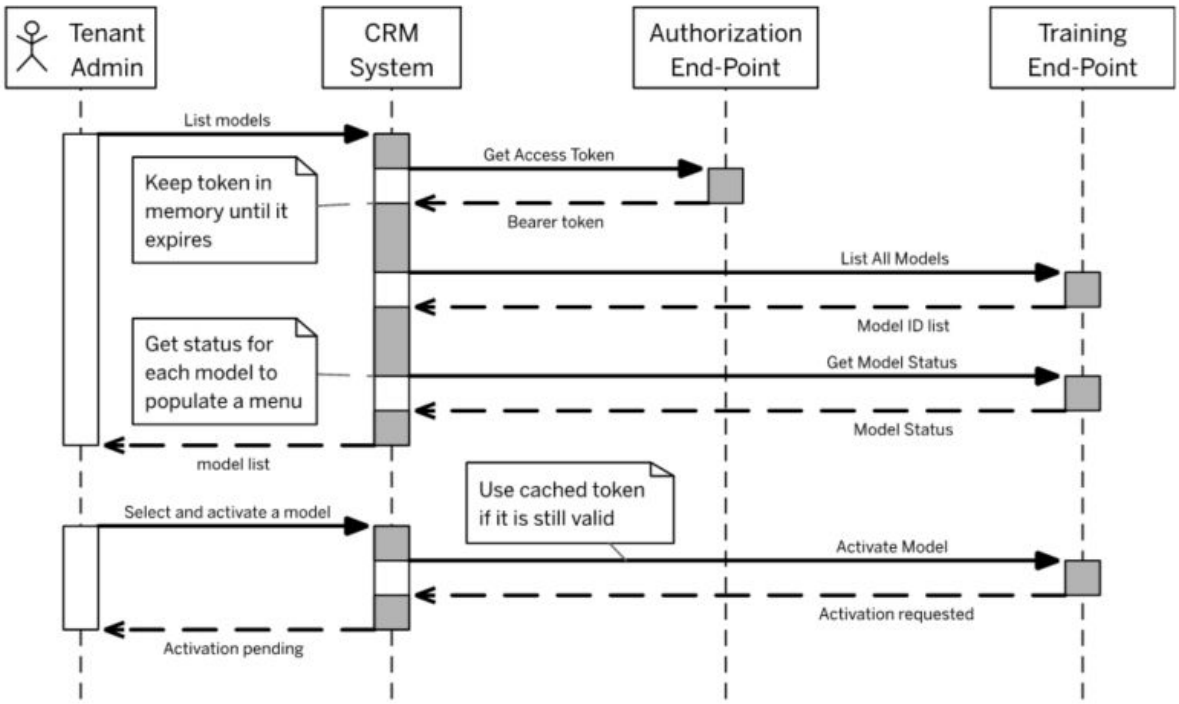
3.2 Model Activation

For a given scenario (language or set of classifications, for example) there can only be one active model. However, the microservice allows you to store more than one model for fallback purposes. The model activation workflow selects an active model for a given scenario. Activating a new model implies that the current active model becomes deactivated.

Related Information

Diagram for Model Activation [page 13]
Get OAuth Token [page 17]
Get Model Status [page 66]
Activate Model [page 85]
List All Models [page 88]

3.2.1 Diagram for Model Activation



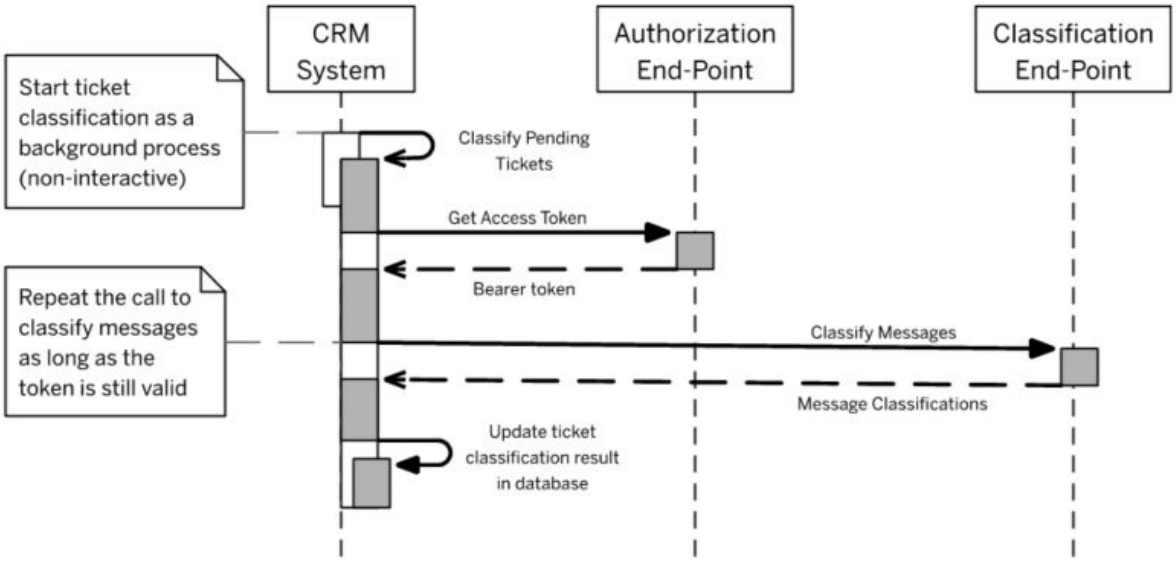
3.3 Classification / Recommendation Inference

The same workflow applies for classification and recommendation inference, it puts a machine learning model to work by predicting how a human would categorize tickets or recommend solutions. This workflow should only be performed on new tickets coming into the CRM system. **Do not invoke the call interactively** (while a user is waiting), but asynchronously or as part of a scheduled job. Moreover, the CRM system may input more than one ticket for a given inference call, limited only by the HTTP payload size.

Related Information

Diagram for Classification / Recommendation Inference [page 14]
Get OAuth Token [page 17]
Classify Messages [page 132]
Recommend Solutions [page 155]

3.3.1 Diagram for Classification / Recommendation Inference



4 Authorization

Related Information

XSUAA Authentication [page 15]

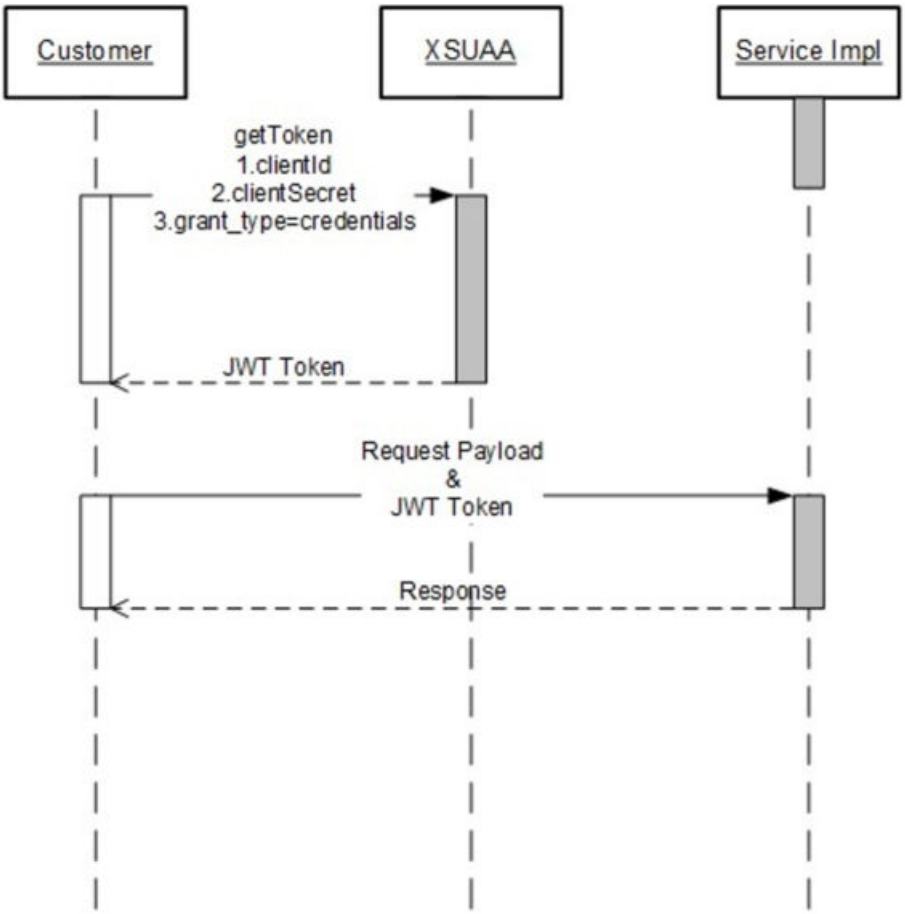
4.1 XSUAA Authentication

Service Ticket Intelligence makes use of the XSUAA Service in SAP Business Technology Platform for authentication and authorization. A valid OAuth Token is required to make calls to the Service Ticket Intelligence API, and the following sections describe how to obtain the OAuth Token.

Related Information

Sequence Diagram to Call Service Ticket Intelligence API [page 16]
Set Up Service in SAP Business Technology Platform [page 16]
Get OAuth Token [page 17]
Input Parameters [page 17]
Expected Response [page 18]
Examples [page 18]

4.1.1 Sequence Diagram to Call Service Ticket Intelligence API



4.1.2 Set Up Service in SAP Business Technology Platform

For the steps to set up Service Ticket Intelligence in SAP Business Technology Platform and more information on this topic, see Set Up Service in SAP Business Technology Platform in the User Guide.

> **i Note**
>
> Parameters from the service key created are required for authorization in Get OAuth Token [page 17].

4.1.3 Get OAuth Token

This request obtains an OAuth Token for use in subsequent calls to Service Ticket Intelligence API. The token is valid for 1799 seconds (30 minutes). After that time, you will need to generate a new one.

The Authorization End-Point is taken from the XSUAA Service URL, which is obtained from the Service Key created to access Service Ticket Intelligence, under the Service Key parameter *uaa-url*. For more information, see Create Service Key in the User Guide.

The request specification is based on the standard Cloud Foundry UAA. For more information, see Cloud Foundry UAA documentation .

Key	Value
End-Point	Authorization
HTTP Method	POST
Path	oauth/token
Headers	Content-Type: application/x-www-form-urlencoded Authorization: Basic <key:secret> encoded in Base64

4.1.4 Input Parameters

This call expects an HTTP Basic authorization header in which the user ID is the UAA clientid and the password is the UAA clientsecret, and the values are obtained from the Service Key created to access Service Ticket Intelligence, under the Service Key parameter uaa-clientid and uaa-clientsecret. For more information, see Create Service Key in the User Guide.

The body should have a URL-encoded form with the following entries:

Parameter	Type	Description
`grant_type`	String	Use value *client_credentials*.

4.1.5 Expected Response

The call returns a JSON dictionary with the following values:

Attribute	Type	Description
`access_token`	String	The bearer token to be used in subsequent calls.
`expires_in`	Integer	The validity of the access token in seconds.

> **i Note**
>
> Ignore other keys in the returned dictionary.

4.1.6 Examples

Example

```
POST /oauth/token HTTP/1.1
Host: sapml-sti.authentication.sap.hana.ondemand.com
Content-Type: application/x-www-form-urlencoded
Authorization: Basic xxxx=
Cache-Control: no-cache
Postman-Token: 21ed81c3-a911-ff3a-ae2a-352fd955dc40
grant_type=client_credentials
```

Example

```
{
    "access_token": "XXXXXXXXX",
    "token_type": "bearer",
    "expires_in": 1799,
    "scope": "ml-sti-dev!b643.sti.all uaa.resource ml-sti-dev!b643.sti.write ml-
sti-dev!b643.sti.standard ml-sti-dev!b643.sti.read",
    "jti": "eec3ec0798224e00ad2451d91e224fa8"
}
```

5 API URL Structure

The URL is a combination of protocol, hostname, product name, version number and end point information in a single string. You must use the URL structure described in the table below to be able to use the following Service Ticket Intelligence APIs:

- Model Training API [page 20]
- Classification API [page 132]
- Recommendation API [page 155]

Component	Description
Protocol	HTTP or HTTPS
Hostname	< sti_service_url > (Retrieved from created Service Key in CF)
Product Name	"sti" for Service Ticket Intelligence
End Point	Respective end-point path

Example

{Protocol}://{Hostname}/{Product Name}/{End Point}

- Protocol: HTTPS
- Hostname: < sti_service_url >
- Product Name: /sti
- End Point: /text/classify

6 Model Training API

Related Information

Upload Training File / Data Extraction [page 20]
Get Model Status [page 66]
Start Model Training [page 83]
Activate Model [page 85]
List All Models [page 88]
Get Model Accuracy [page 96]
Delete Model or Training File [page 105]
Get Training Topics Tabulated [page 107]
Upload Incremental Training File / Data Extraction [page 113]
Disable Model Category [page 122]
Enable Model Category [page 125]
Deactivate Model [page 129]

6.1 Upload Training File / Data Extraction

Upload a training file to create new ML models. The ML configuration settings contain a set of four items: scenario, field mapping, topic tabulating, and training settings. Uploading a training file results in a model ID to be used to create its corresponding machine learning model.

Each subscriber may keep a maximum of 20 training models. Once this limit is reached, further upload requests will be denied, until one of the existing models is removed using the endpoint in Delete Model or Training File [page 105].

For better inference result in Classification scenario type, the recommended number of records is at least 1,000 samples for every unique label value (for example a combination of output fields and each field's unique values). For example, in a simple binary classifier of "good" and "bad" classes, there should be at least 1000 samples of "good" text and another 1,000 samples of "bad" text. Moreover, a minimum of 20 records are required for each unique label value – records belonging to categories that do not meet this minimum threshold are treated as outlier and removed.

The Upload Training File endpoint also provides Data Extraction functionality. This triggers data extraction from the provided OData service asynchronously and results in a model ID to be used to create its corresponding machine learning model.

For data extraction, OData information should be provided in the request instead of the file content. The data extraction request can have the following variations:

- With template: the basic predefined OData query is used to retrieve the data.
- Without template: the Data extractor request should contain all the necessary information to construct the OData request.

The status of the model can be retrieved or monitored from the Get Model Status endpoint. See Get Model Status [page 66]. Initially the model status is set to DATA_EXTRACTION_IN_PROGRESS. Once the data extraction is complete, the model status changes to NEW. If the data extraction failed, the model status is updated to DATA_EXTRACTION_FAILED and you can see the failure cause.

Key	Value
End-Point	Training
HTTP Method	POST
Path	Model
Headers	Content-Type: application/json Authorization: Bearer <token> X-Request-ID: <unique string>

Related Information

Input Attributes [page 21]
Expected Response [page 36]
Examples [page 43]

6.1.1 Input Attributes

The body should contain a JSON dictionary with the following entries:

Attribute	Type	Description
`scenario`	Scenario Object	Scenario details: scenario name, type, language, business object. See Scenario Object [page 23].
`mapping`	Mapping Object	Fields you use as input for training. See Mapping Object [page 24].

Attribute	Type	Description
`tabulating`	Tabulating Object (optional)	Fields you use as source and topics for topic tabulation. See Tabulating Object [page 28].
`options`	Options Object (optional)	Allows you to perform additional file operations during file upload. See Options Object [page 29].
`training`	Training Object (optional)	Uploads the file used as training data. Use this object only for file upload functionality. For data extraction, this object will be ignored. See Training Object [page 32].
`odata`	OData Object (optional)	OData information you use for data extraction. Use this object only for data extraction functionality. See OData Object [page 32].
`transformations`	Array of key, value pairs (optional)	Fields that will be used during OData extraction to remap or remove categories. Use this object only for data extraction functionality. See Transformations Object [page 35].

Related Information

Scenario Object [page 23]
Mapping Object [page 24]
Tabulating Object [page 28]
Options Object [page 29]
Training Object [page 32]
OData Object [page 32]
Transformations Object [page 35]

6.1.1.1 Scenario Object

The Scenario object name in JSON is `scenario`.

Attribute	Type	Description
`desc`	String (optional)	A short description about the contents of the training file.
`type`	String	It specifies the type of Machine Learning service to be trained against. Currently the possible types are: • Classification: text classification The ML algorithm learns from historical data (such as service tickets with unstructured messages with the corresponding categories), generates an ML model and uses it to predict output categories for new incoming service tickets. See Classification API [page 132] for inference definitions. • Recommendation: solution recommendation The ML algorithm maps user queries (incoming service tickets) and/or answers KB articles into a latent semantic space, and returns output recommendations with most relevant queries and/or answers for new incoming service tickets. See Recommendation API [page 155] for inference definitions. • Clustering: text clustering The ML algorithm uses both TF-IDF and word embedding to learn the representation of tickets in an unsupervised manner. Gain insights from unstructured data to identify trends or patterns in tickets from the clusters and keywords that are identified. See Clustering API [page 165] for inference definitions.

Attribute	Type	Description
`language`	String (optional)	It denotes the language of the training file. In case the language is not specified, English is assumed. For the recommendation scenario type, if the dataset is a mix of multiple languages, then the same will be automatically evaluated during training. The user can make a call to Get Model Status [page 66] after training to check the language mix. For an example of sample response for Get model status for scenario type recommendation with language mix, see Get Model Status for Recommendation with Language Mix [page 82]. See the list of supported languages and their codes in Supported Languages.
`business_object`	String	It specifies the business object of the model being trained. There can be only one ACTIVE model per scenario, per business object and per language. Currently only *ticket* is valid.

6.1.1.2 Mapping Object

A Mapping object describes which columns in the upload file should be used as sample input and which ones are to be used as the classification or recommendation output that the model should learn.

Input columns can be specified simply as an array of strings, in which case the data type for all input columns are interpreted as string by default.

For scenario type Classification, input columns may alternatively be specified as an array of Field-Type Object with individual data type (str/num/cat) specified for each input column. Output columns for scenario type Classification are categorical by definition.

For scenario type Recommendation, input and output columns are always interpreted as type string. Array of Field-Type Object is not applicable in this case.

The Mapping object name in JSON is `mapping`.

Attribute	Type	Description
`input`	Array<String>	A collection of strings referring to the column names in the upload file that are to be considered as sample data in the classification or recommendation problem.
	Array of Field-Type Object (applicable for scenario type Classification only)	A collection of Field-Type objects referring to the column name and its type in the upload file, to be considered as sample data in the classification problem. See Field-Type Object [page 27].
`output`	Array<String>	A collection of strings referring to the column names in the upload file denoting the model outputs. **i Note** When performing Recommendation model type training, it is possible to specify the answer-KB article as the model output. This however, means that the original answer-KB article is copied and stored as part of the trained model. Hence, it is strictly prohibited for users that do not have copyright ownership on respective KB article.

Attribute	Type	Description
`priority`	String (optional)	Optional field used to indicate which column should be considered the `priority` field for tickets. For example, the priority can be high, low or medium, or can be numeric (0, 1, 2, 3 and so on). The field specified as priority must be present as one of the output fields while submitting the OData extraction request. **i Note** The `priority` field cannot be used for file upload and currently it only supports English and German. See Data Extractor with Priority for Priority Type Training [page 57] for a sample code example.
`keywords`	Array of dictionaries <columns, text and value being the keys> (optional)	Optional field used to map the `priority` field to a certain category. For example, if the input field(s) contain certain word or words then the priority should be **high** instead of low or medium. If a ticket is marked as low priority and it contains certain text indicating that it should be a **high** priority ticket, the relabeling feature fixes the priority as part of data cleaning. Alternatively, if a ticket is marked as **high**, but in fact the priority should be low or medium based on a certain text in the ticket, the relabeling feature can also be used. See Data Extractor with Priority and Keywords (Relabeling) for Priority Type Training [page 57] for a sample code example.

Attribute	Type	Description
`groupby`	String (optional)	The `groupby` field selects an input field in which the possible values on the corresponding output field(s) would be restricted by the associated value on the `groupby` field, as represented in the training data. This is useful to restrict predictions based on single-level groupings of the input. In a fast-food scenario, for example, the input fields are **`category`**, **`message text`** and the output field is **`menu item`**. The **`category`** input field may contain either burgers or drinks whereas the **`menu item`** output field will have cheese burger, fish fillet, veggie patty, coffee, tea, or soda. Hence, the model would predict which menu item the customer is ordering given free-form text in the **`message text`** and a pre-selected value in the **`category`** field. Now the `groupby` field is set as the **`category`** input field. Since the training data never associates burgers input to soda or coffee output, the model would similarly never output soda or coffee when the **`category`** field is burgers. See Upload File with Groupby for Classification Type Training [page 59] for a sample code example.

Related Information

Field-Type Object [page 27]

6.1.1.2.1 Field-Type Object

A Field-Type object provides the name and type for a field in the training data.

The Field-Type object has the following names in JSON:

- `input` (Array)
- `extracted_input` (Array)

Attribute	Type	Description
`field`	String	The strings refer to the column names in the upload file that are to be considered as sample data in the classification problem.
`type`	Integer	It denotes the input type of the field column in the upload file. Possible values are: • str: string input • num: numeric input • cat: categorical input

See Upload File with Field-Type Object for Classification Type Training [page 44] for a sample code example.

> **i Note**
>
> The `extracted_input` (Array) attribute belongs to the Model Status object. For more information, see Model Status Object [page 37].

6.1.1.3 Tabulating Object

A list of tabulation objects describes which columns in the upload file should be used as sample input source and maps them with input topic for the purpose of token tabulation. The list is an optional object and a maximum of three tabulation objects is allowed. If specified, then topic-tabulation results are available as part of the model. Currently Tabulating Object is valid only for the scenario type Classification.

The Tabulating object name in JSON is `tabulating`.

Attribute	Type	Description
`source`	String	The string referring to the column name in the upload file that is to be considered as sample data for topic tabulation.
`topic`	String	The string referring to the column name in the upload file that suggests the tabulation topic.

6.1.1.4 Options Object

The options object allows you to perform additional file operations during file upload.

The Upload Training File Options object name in JSON is `options`.

Attribute	Type	Description
`anonymization`	String (optional)	Anonymizes common names in training data during pre-processing to mask sensitive information. Currently, the `anonymization` attribute is valid only for the scenario type Classification.
`bulk_file_upload`	Boolean (optional)	Allows you to upload multiple files during file upload. Each file part should not exceed the file size limit (file upload size) in Technical Constraints. See examples in Bulk File Upload [page 62].
`bulk_file_upload_complete`	Boolean (optional)	Specify this attribute on the final file part upload to complete the entire multiple file upload operation. By default, this value is set to false. See examples in Bulk File Upload [page 62].
`encoding`	String (optional)	Specifies the codec that the file is encoded in. By default, it is assumed that the file is in UTF-8 encoding. Currently, the `encoding` attribute is valid only for the scenario type Classification. See Supported Encodings [page 30].

Attribute	Type	Description
`lang_config`	List (optional)	List of key or value pairs. Key is the language and value is a list of stop words for that language. Currently, the `lang_config` attribute is valid for the scenario types Recommendation and Clustering. See an example in Upload File for Recommendation with Stop Words [page 48].
`training_file_field_delimiter`	String (optional)	Specifies an alternative delimiter character for the file upload CSV file. When not specified, the default is the standard comma "," character. Currently, the `training_file_field_delimiter` attribute is valid only for the scenario type Classification. See an example in Upload File with Custom Field Delimiter for Classification Type Training [page 50].

6.1.1.4.1 Supported Encodings

Service Ticket Intelligence supports the following standard encodings:

> → Tip
>
> UTF-8 encoding is highly recommended.

Codec	Aliases	Languages
`UTF_8`	U8, UTF, utf8, UTF-8	all languages
`UTF_8_SIG`		all languages
`ASCII`	us-ascii	English
`BIG5`	big5-tw	Traditional Chinese

Codec	Aliases	Languages
GB2312	chinese, csiso58gb231280, euc-cn, euccn, eucgb2312-cn, gb2312-1980, gb2312-80, iso-ir-58	Simplified Chinese
GB18030	gb18030-2000	Unified Chinese
HZ-GB-2312	hzgb, hz-gb, hz	Simplified Chinese
ISO2022_JP	csiso2022jp, iso2022jp, iso-2022-jp	Japanese
CP932	932, ms932, mskanji, ms-kanji	Japanese
EUC-JP	eucjp, ujis, u-jis	Japanese
SHIFT_JIS	csshiftjis, shiftjis, sjis, s_jis	Japanese
EUC-KR	euckr, korean, ksc5601, ks_c-5601, ks_c-5601-1987, ksx1001, ks_x-1001	Korean
ISO2022_KR	csiso2022kr, iso2022kr, iso-2022-kr	Korean
KOI8_R		Russian
MACCYRILLIC	mac_cyrillic	Bulgarian, Byelorussian, Macedonian, Russian, Serbian
IBM855	cp855, 855	Bulgarian, Byelorussian, Macedonian, Russian, Serbian
IBM866	cp866, 866	Russian
ISO-8859-2	iso8859_2	Central and Eastern Europe
ISO-8859-5	cyrillic, iso8859_5	Bulgarian, Byelorussian, Macedonian, Russian, Serbian
WINDOWS-1251	cp1251	Bulgarian, Byelorussian, Macedonian, Russian, Serbian
ISO-8859-1	iso8859-1, 8859, cp819, latin, latin-1	Western Europe
WINDOWS-1252	cp1252	Western Europe
ISO-8859-7	iso8859_7, greek, greek8	Greek
WINDOWS-1253	cp1253	Greek
ISO-8859-8	iso8859_8, hebrew	Hebrew
WINDOWS-1255	cp1255	Hebrew
UTF-16	utf_16	all languages

Codec	Aliases	Languages
`UTF-32`	utf_32	all languages

6.1.1.5 Training Object

The training object contains the training file in a CSV format with a header row. This header row **must** contain *input* and *output* columns as described in Mapping Object [page 24]. The header row can also be specified as *source* and *topic* columns as described in Tabulating Object [page 28].

The Training object name in JSON is `training`.

Attribute	Type	Description
`file`	String	It uploads the file encoded as a Base64 string. **i Note** Only CSV files are supported.

6.1.1.6 OData Object

The OData object name in JSON is `odata`.

Attribute	Type	Description
application	String (optional)	A short name of the approved calling application (C4C, for example). When this field is not present, the default OData client would be used.
credential	Credential Object	OData information to be used for data extraction. See Credential Object [page 33].
query_options	Query Options Object (optional)	The query options to construct the OData data retrieval query. Either *template* or *query_options* can be provided. See Query Options Object [page 34].

Related Information

Credential Object [page 33]
Query Options Object [page 34]

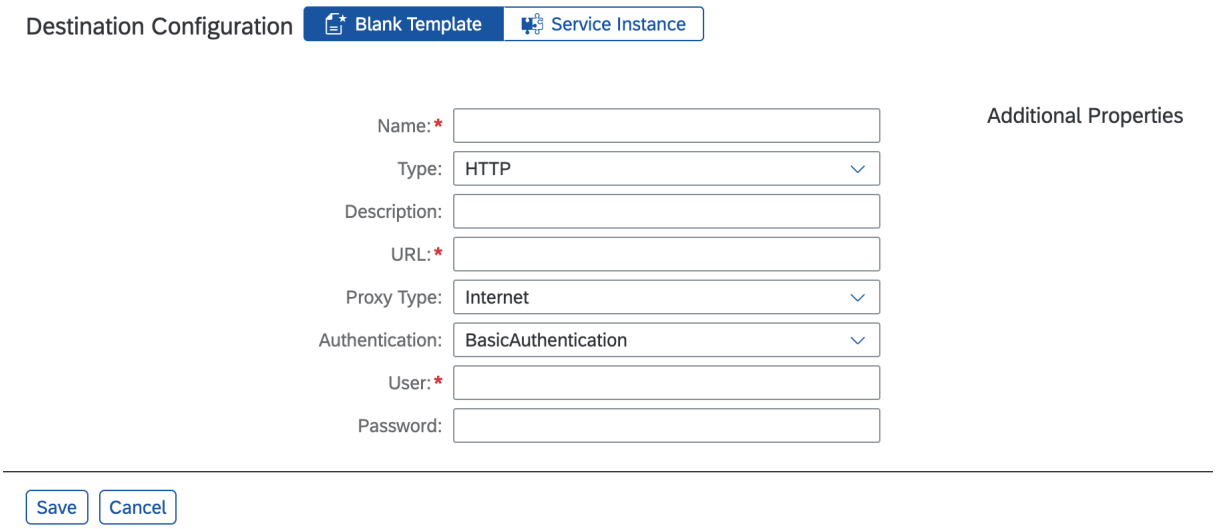
6.1.1.6.1 Credential Object

The Credential object name in JSON is `credential`.

Attribute	Type	Description
service_url	String	A valid URL to OData service.
authorization ⚠ Caution This attribute is deprecated since 18.12.2020.	String	A valid Basic or Bearer authorization token for the provided OData service.

Destination Service Support

Service Ticket Intelligence allows *Destination Configuration* for OData extraction. For that, you need to create a *New Destination* on your Cloud Foundry subaccount.



1. Enter a *Name* of your choice.

2. Enter *URL*, *User* and *Password*.
3. *Save* your *Destination Configuration*.

Once destination is created, you need to make one change to the file upload request for initiating OData extractions. Instead of specifying an HTTPS URL scheme, use the “x-sap-destination” URI scheme. So, for example, if *Name* is “sample”, the same can be passed in the OData request body in the following format: "service_url": "x-sap-destination:sample".

See Upload File with Destination Configuration for OData Extraction [page 56] for a sample code example.

The rest of the request stays as per the usual OData extraction. Service Ticket Intelligence gets the destination details and use them to proceed with the extraction.

Related Information

Consuming the Destination Service

6.1.1.6.2 Query Options Object

A Query Options Object describes the entity, selects and filters conditions as per OData standard.

The Query Options object name in JSON is `query_options`.

Attribute	Type	Description
entity	String	A valid entity name of provided OData service.
select	String	Column names (separated by commas) in the provided entity and/or expanded entity that needs to be retrieved.
filter	String (optional)	Valid filter conditions that need to be applied on retrieval.
orderby	String (optional)	Valid order by condition that needs to be applied on retrieval.
expand	String (optional)	Entity names (separated by commas) that need to be expanded on retrieval.
batch_query_options	Batch Query Options Object (optional)	Child records that need to be retrieved as OData batch call. See Batch Query Options Object [page 35].

Related Information

Batch Query Options Object [page 35]

6.1.1.6.2.1 Batch Query Options Object

A Batch Query Options Object has the same structure as Query Options Object. Child records of Query Options Object can be retrieved as OData batch call if the Batch Query Options Object is provided.

The Batch Query Options object name in JSON is `batch_query_options`.

Attribute	Type	Description
entity	String	A valid child entity name of provided entity.
select	String	Column names (separated by commas) in the provided entity and/or expanded entity that needs to be retrieved.
filter	String (optional)	Valid filter conditions that need to be applied on retrieval.
orderby	String (optional)	Valid order by condition that needs to be applied on retrieval.
expand	String (optional)	Entity names (separated by commas) that need to be expanded on retrieval.

6.1.1.7 Transformations Object

This is an optional feature which is only applicable for OData file upload and will be used for remapping or removing some of the categories. Service Ticket Intelligence reads this transformation JSON and does the preprocessing of the data before training happens. Currently only replace and remove operations are supported. See Sample Request (with Transformations Object) example in Data Extractor for Classification Type Training [page 51].

The Transformations object name in JSON is `transformations`.

Attribute	Type	Description
`output_category`	String	Name referring to the output column in the upload file generated as part of OData extraction.
`operation`	Key, value pairs	`operation` is a map with keys such as `operation_type` and `rules`.
`operation_type`	String	The type of the operation. Currently, only replace and remove are supported.
`rules`	Array of key, value pairs	Each mapping specifies the old value and new value.
`category_class`	String	Old value.
`new_category_class`	String	New value. For remove operation it will be an empty string, for example, `new_category_class`: " ".

6.1.1.8 Bulk File Upload Object

Attribute	Type	Description
complete	Boolean	Indicates if the file upload is complete.
file_count	Integer	The total number of files currently uploaded.

Related Information

Bulk File Upload [page 62]

6.1.2 Expected Response

The call returns a model status object (similar to Get Model Status [page 66]) containing the model identifier and summary of the training data for file upload functionality. See Model Status Object [page 37].

> **i Note**
>
> The same fields will be returned in the response body of Upload Training File / Data Extraction [page 20] and Get Model Status [page 66].

6.1.2.1 Model Status Object

Attribute	Type	Description
`bulk_file_upload`	Dictionary	Multiple file upload details. See Options Object [page 29].
`business_object`	String	The type of business object to classify. This was specified as part of the training scenario. See Scenario Object [page 23].
`combined_accuracy` **i Note** Applicable for scenario type classification only.	Float	The expected accuracy of the classification model, for all output fields. This is returned only when the model has been successfully trained.
`data_extraction_end_timestamp`	String	The UTC time stamp captured in the end of data extraction request (return to NEW model status). Format '%Y-%m-%d %H:%M:%S UTC'
`data_extraction_failure_cause`	String	Error message captured during the data extraction failed.
`data_extraction_start_timestamp`	String	The UTC time stamp captured during start of data extraction request. Format '%Y-%m-%d %H:%M:%S UTC'
`desc`	String	The short description as specified during training file upload. See Scenario Object [page 23].
`disabled_category_class`	Array of Value field-categories objects	The category class disabled from model. See Value Field-Categories Object [page 42] for definition.

Attribute	Type	Description
`enabled_category_class`	Array of Value field-categories objects	The category class which is enabled in model, only an enabled category class is eligible for inference. See Value Field-Categories Object [page 42] for definition.
`extracted_input`	Array of Input Fields	Provides a list of input mapping fields. See Field-Type Object [page 27].
`extracted_output`	Array of Field Summary objects	Provides a histogram of each output field. See Field Summary Object [page 41].
`file_upload_timestamp`	String	The UTC time stamp captured during upload training file. Format '%Y-%m-%d %H:%M:%S UTC'
`incremental`	Boolean	The binary indicator for initial or incremental training file upload as follows: • False: fresh training file upload (default) • True: incremental training file upload
`incremental_possible`	Boolean	The binary indicator for whether the model is eligible as reference model for incremental file upload, or the model cannot be used as reference model as follows: • True: incremental is possible, the model version is compatible with runtime version • False: incremental is **not** possible, the model version is not compatible **i Note** This attribute is ony available in trained models.
`incremental_upload_timestamp`	String	The UTC time stamp captured during upload incremental training file. Format '%Y-%m-%d %H:%M:%S UTC'

Attribute	Type	Description
`language`	String	The language code as specified during training file upload. See Scenario Object [page 23].
`language_mix` **i Note** Applicable for scenario type recommendation only. It appears only after training if there is a mix of languages in the dataset.	Array of String	Describes the mix of languages found in the dataset. For example: `"language_mix": [ "en", "de", "es" ],`
`model_activation_timestamp`	String	The UTC time stamp captured during active model process. Format '%Y-%m-%d %H:%M:%S UTC'
`model_deactivate_timestamp`	String	The UTC time stamp captured when an activate model is deactivated (return to READY model status). Format '%Y-%m-%d %H:%M:%S UTC'
`model_id`	String	Unique identifier for the model.
`model_status`	String Enumeration	Status of the model. See Model Status [page 7] for model status values.
`new_category_class`	Array of Value field-categories objects	The new category class found in the training set during incremental training file upload. See Value Field-Categories Object [page 42] for definition.
`progress_level`	Integer	Percentage progress of an on-going training or data extraction process. The value is *100* upon completion.

Attribute	Type	Description
`recall` **i Note** Applicable for scenario type recommendation only. It appears only after training if there is a mix of languages in the dataset.	Array of Value language-recall_rate objects	Indicates how likely the correct language results are described by the `language_mix` attribute. Recall value varies between 1 to 100. For example, if the dataset comprises of 3 languages, the accuracy is evaluated by splitting the dataset as per language into 3 parts: `"Recall": [ { "language": "en", "recall_rate": 100 }, { "language": "de", "recall_rate": 100 }, { "language": "es", "recall_rate": 90 } ],`
`recall@k` (k is an integer, for example 5,10) **i Note** Applicable for scenario type recommendation only.	Float	Indicates how likely the correct solution is within the top-k recommendation results. For example if the correct solution is always present in top-k then recall@k will be 100. Recall value varies between 1 to 100.
`records_count`	Integer	The number of records in the uploaded training file. **i Note** In case of data extraction, the `records_count` attribute refers to the number of records extracted successfully.

Attribute	Type	Description
`reference_model_id`	String	The unique identifier for the reference model for incremental file upload. Only applicable for incremental training file upload. See Input Attributes [page 114].
`status`	Integer	Function call status code. See Common Status and Error Codes [page 171] for possible values.
`status_message`	String	User-friendly error message.
`tabulations_count`	Integer	The number of tabulation objects specified during training file upload. See Tabulating Object [page 28].
`total_record_count`	Integer	The total number of records to be retrieved during data extraction.
`training_finish_timestamp`	String	The UTC time stamp captured. Format '%Y-%m-%d %H:%M:%S UTC'
`training_start_timestamp`	String	The UTC time stamp captured during start model training process. Format '%Y-%m-%d %H:%M:%S UTC'
`type`	String	Specifies the type of Machine Learning service as specified during training file upload. See Scenario Object [page 23].

Related Information

Field Summary Object [page 41]
Value Field-Categories Object [page 42]

6.1.2.1.1 Field Summary Object

A Field Summary object provides a summary about a field in the training data.

The Field Summary object name in JSON is `extracted_output` (Array).

Attribute	Type	Description
`field`	String	Unique identifier for the model.
`distribution` **i Note** Applicable for scenario type Classification only.	Array of Value Distribution objects	Provides a histogram of every value occurring in the training data. See Value Distribution Object [page 42] for definition.

Related Information

Value Distribution Object [page 42]

6.1.2.1.1.1 Value Distribution Object

A Value Distribution object shows how many times the value occurs within a training set.

The Value Distribution object name in JSON is `distribution` (Array).

Attribute	Type	Description
`value`	String	The enumeration value within a field of the training set.
`count`	Integer	The number of times the associated value occurs.

6.1.2.1.2 Value Field-Categories Object

A Value Field-Categories object shows the category class value occurs within a training set.

The Value Field-Categories object has the following names in JSON:

- `new_category_class` (Array)
- `enabled_category_class` (Array)
- `disabled_category_class` (Array)

Attribute	Type	Description
`field`	String	The name of the field. This needs to match one of the field names given in the `output` attribute in Mapping Object [page 24].
`category_class`	Array of Category Class value	The enumeration value within a field of the training set.

6.1.3 Examples

Related Information

Upload File for Classification Type Training [page 43]
Upload File with Field-Type Object for Classification Type Training [page 44]
Upload File for Recommendation Type Training [page 46]
Upload File for Recommendation with Stop Words [page 48]
Upload File with Custom Field Delimiter for Classification Type Training [page 50]
Data Extractor for Classification Type Training [page 51]
Data Extractor with Priority for Priority Type Training [page 57]
Data Extractor with Priority and Keywords (Relabeling) for Priority Type Training [page 57]
Upload File with Groupby for Classification Type Training [page 59]
Upload File with Encoding Option for Classification Type Training [page 60]
Upload File for Clustering Type Training [page 61]

6.1.3.1 Upload File for Classification Type Training

Example

Sample Training Data and Format

```
subject,body,ticket_type,routing_inbox,
"jacket found","Massive thanks for finding my jacket (aka the best travel pillow/
blanket/lumbar support ever)!",compliment,public_relations,
"kudos!","shoutout to flight attendant Angela Baby 5566 SIN-LAX yesterday. She
handled my daughter’s motion sickness like a pro.",compliment,public_relations,
```

```
"thanks!","Love getting unexpected upgrade for the ride home.
Thanks",compliment,public_relations
```

Example

Sample Request


```
{
  "scenario": {
    "desc": "Simple Twitter Training Data",
    "type": "classification",
    "language": "en",
    "business_object": "ticket"
  },
  "mapping": {
    "input": ["subject", "body"],
    "output": ["ticket_type", "routing_inbox"]
  },
  "tabulating": [
    {
      "source": "body",
      "topic": "ticket_type"
    },
    {
      "source": "body",
      "topic": "routing_inbox"
    }
  ],
  "training": {
    "file":
"c3ViamVjdCxib2R5LHRpY2tldF90eXBlLHJvdXRpbmdfaW5ib3gsCiJqYWNrZXQgZm91bmQiLCJNYXNz
aXZlIHRoYW5rcyBmb3IgZmluZGluZyBteSBqYWNrZXQgKGFrYSB0aGUgYmVzdCB0cmF2ZWwgcGlsbG93L
2JsYW5rZXQvbHVtYmFyIHN1cHBvcnQgZXZlcikhIixjb21wbGltZW50LHB1YmxpY19yZWxhdGlvbnMsCi
JrdWRvcyEiLCJzaG91dG91dCB0byBmbGlnaHQgYXR0ZW5kYW50IEFuZ2VsYSBCYWJ5IDU1NjYgU0lOLUx
BWCB5ZXN0ZXJkYXkuIFNoZSBoYW5kbGVkIG15IGRhdWdodGVy4oCZcyBtb3Rpb24gc2lja25lc3MgbGlr
ZSBhIHByby4iLGNvbXBsaW1lbnQscHVibGljX3JlbGF0aW9ucywKInRoYW5rcyEiLCJMb3ZlIGdldHRpb
mcgdW5leHBlY3RlZCB1cGdyYWRlIGZvciB0aGUgcmlkZSBob21lLiBUaGFua3MiLGNvbXBsaW1lbnQscH
VibGljX3JlbGF0aW9ucwo="
  }
}
```


6.1.3.2 Upload File with Field-Type Object for Classification Type Training

Example

Sample Request


```
{
  "scenario": {
    "desc": "Simple Twitter Training Data",
```

```
    "type": "classification",
    "language": "en",
    "business_object": "ticket",
  },
  "mapping": {
    "input": [
      {
        "type": "str",
        "field": "subject"
      },
      {
        "type": "str",
        "field": "body"
      }
    ],
    "output": [
      "ticket_type",
      "routing_inbox"
    ]
  },
  "tabulating": [
    {
      "source": "body",
      "topic": "ticket_type"
    },
    {
      "source": "body",
      "topic": "routing_inbox"
    }
  ],
  "training": {
    "file":
"c3ViamVjdCxib2R5LHRpY2tldF90eXBlLHJvdXRpbmdfaW5ib3gKamFja2V0IGZvdW5kLE1hc3NpdmUg
dGhhbmtzIHRvICBmb3IgZmluZGluZyBteSAgamFja2V0IChha2EgdGhlIGJlc3QgdHJhdmVsIHBpbGxvd
y9ibGFua2V0L2x1bWJhciBzdXBwb3J0IGV2ZXIpISxjb21wbGltZW50LHB1YmxpY19yZWxhdGlvbnMKa3
Vkb3MhLHNob3V0b3V0IHRvIGZsaWdodCBhdHRlbmRhbnQgQW5nZWxhIEFBIDI1NSBKRkstTEFYIHllc3R
lcmRheS4gU2hlIGhhbmRsZWQgbXkgZGF1Z2h0ZXJzIG1vdGlvbiBzaWNrbmVzcyBsaWtlIGEgcHJvLixj
b21wbGltZW50LHB1YmxpY19yZWxhdGlvbnMKdGhhbmtzISxMb3ZlIGdldHRpbmcgdW5leHBlY3RlZCB1c
GdyYWRlIGZvciB0aGUgcmlkZSBob21lLiBUaGFua3MgIChhdCAgICxjb21wbGltZW50LHB1YmxpY19yZW
xhdGlvbnM="
  }
}
```


Example

Sample Response


```
{
  "model_id": "39984FA8EE524AFAA0E4BF3860E31867",
  "model_status": "NEW",
  "desc": "Simple Twitter Training Data",
  "type": "classification",
  "business_object": "ticket",
  "language": "en",
  "tabulations_count": 2,
  "records_count": 3,
  "incremental": false,
  "extracted_input": [
    {
      "type": "str",
      "field": "subject"
    },
    {
```

```
      "type": "str",
      "field": "body"
    }
  ],
  "extracted_output": [
    {
      "field": "ticket_type",
      "distribution": [{
          "value": "compliment", "count": 3
        }]
    },
    {
      "field": "routing_inbox",
      "distribution": [{
          "value": "public_relations", "count": 3
        }]
    }
  ],
  "status": 0,
  "status_message": "ok"
}
```

6.1.3.3 Upload File for Recommendation Type Training

Example

Sample Training Data and Format

```
subject,body,data_source,article_id,
"How to find the serial number?","How to find the serial
number?","mindtouch",8FF89FC8983134C863729AC5B891D238"
"How to adjust smartouch display?","How to adjust smartouch
display?","mindtouch","F3DDBF5D5CC83D17C509E734EBA215D6"
"How to check leave balance?","How to check leave
balance?","mindtouch","46F04364AB0034953C0C7BC594B1B3CE"
```

Example

Sample Request

```
{
  "scenario": {
    "desc": "Simple Question Recommendation Training Data",
    "type": "recommendation",
    "language": "en",
    "business_object": "ticket",
  },
  "mapping": {
    "input": ["subject", "body"],
    "output": ["data_source", "article_id"]
  },
```

```
  "training": {
    "file": "
c3ViamVjdCxib2R5LGRhdGFfc291cmNlLGFydGljbGVfaWQsDQoiSG93IHRvIGZpbmQgdGhlIHNlcmlhb
CBudW1iZXI/IiwiSG93IHRvIGZpbmQgdGhlIHNlcmlhbCBudW1iZXI/
IiwibWluZHRvdWNoIiw4RkY4OUZDODk4MzEzNEM4NjM3MjlBQzVCODkxRDIzOCINCiJIb3cgdG8gYWRqd
XN0IHNtYXJ0b3VjaCBkaXNwbGF5PyIsIkhvdyB0byBhZGp1c3Qgc21hcnRvdWNoIGRpc3BsYXk/
IiwibWluZHRvdWNoIiwiRjNEREJGNUQ1Q0M4M0QxN0M1MDlFNzM0RUJBMjE1RDYiDQoiSG93IHRvIGNoZ
WNrIGxlYXZlIGJhbGFuY2U/IiwiSG93IHRvIGNoZWNrIGxlYXZlIGJhbGFuY2U/
IiwibWluZHRvdWNoIiwiNDZGMDQzNjRBQjAwMzQ5NTNDMEM3QkM1OTRCMUIzQ0Ui"
  }
}
```


Example

Sample Response


```
{
  "model_id": "39984FA8EE524AFAA0E4BF2360E31867",
  "model_status": "NEW",
  "desc": "Simple Question Recommendation Training Data",
  "type": "recommendation",
  "business_object": "ticket",
  "language": "en",
  "records_count": 3,
  "incremental": false,
  "extracted_input": [
    {
      "type": "str",
      "field": "subject"
    },
    {
      "type": "str",
      "field": "body"
    }
  ],
  "extracted_output": [
    {
      "field": "data_source"
    },
    {
      "field": "article_id"
    }
  ],
  "status": 0,
  "status_message": "ok"
}
```

6.1.3.4 Upload File for Recommendation with Stop Words

Example

Sample Request


```
{
   "scenario":{
      "desc":"Train Similar Tickets with random tickets (small)",
      "type":"recommendation",
      "language":"en",
      "business_object":"ticket"
   },
   "mapping":{
      "input":[
         "subject",
         "body"
      ],
      "output":[
         "article_desc",
         "article_id",
         "datasource",
         "article_url"
      ]
   },
   "training":{

"file":"c3ViamVjdCxib2R5LGFydGljbGVfaWQsYXJ0aWNsZV9kZXNjLGRhdGFzb3VyY2UsYXJ0aWNsZ
V91cmwKIkknbSBhIHRyaXBsZSBDYXByaWNvcm4gKFN1biwgTW9vbiBhbmQgYXNjZW5kYW50IGluIENhcH
JpY29ybikgV2hhdCBkb2VzIHRoaXMgc2F5IGFib3V0IG1lPyIsLDEscmVsaWdpb3VzIHNlbnRpbWVudHM
sbWluZHRvdWNoLHd3dy50aW1lc29maW5kaWEuY29tCldoYXQgc2hvdWxkIEkgZG8gdG8gYmUgYSBncmVh
dCBnZW9sb2dpc3Q"
   },
   "options":{
      "lang_config":[
         {
            "language":"en",
            "stopwords":[
               "eng1stopword",
               "eng2stopword",
               "eng3stopword",
               "sanity",
               "STIDemo",
               "SAPERP",
               "Dummy"
            ]
         },
         {
            "language":"de",
            "stopwords":[
               "welcher",
               "testingdestopword",
               "daran",
               "rechter",
               "unserer",
               "musste"
            ]
         }
      ]
   }
```

```
}
```

Example

Sample Response


```
{
   "business_object":"ticket",
   "desc":"Train Answer Recommendation random questions (small)",
   "disabled_category_class":[

   ],
   "enabled_category_class":[

   ],
   "extracted_input":[
      {
         "field":"subject",
         "type":"str"
      },
      {
         "field":"body",
         "type":"str"
      }
   ],
   "extracted_output":[
      {
         "field":"article_desc"
      },
      {
         "field":"article_id"
      },
      {
         "field":"datasource"
      },
      {
         "field":"article_url"
      }
   ],
   "file_upload_timestamp":"2020-10-17 07:02:44 UTC",
   "incremental":false,
   "incremental_possible":true,
   "language":"en",
   "model_id":"603e4194a4f24aa7a1aebe17d095e526",
   "model_status":"NEW",
   "new_category_class":[

   ],
   "records_count":2600,
   "status":0,
   "status_message":"ok",
   "tabulations_count":0,
   "type":"recommendation"
}
```


> **i Note**
>
> The uploaded stop words can only be seen after training gets completed.

6.1.3.5 Upload File with Custom Field Delimiter for Classification Type Training

Example

Sample Training Data and Format

```
subject;body;ticket_type;routing_inbox;
"jacket found";"Massive thanks for finding my jacket (aka the best travel pillow/
blanket/lumbar support ever)!";compliment;public_relations;
"kudos!";"shoutout to flight attendant Angela Baby 5566 SIN-LAX yesterday. She
handled my daughter’s motion sickness like a pro.";compliment;public_relations;
"thanks!";"Love getting unexpected upgrade for the ride home.
Thanks";compliment;public_relations
```

Example

Sample Request

```
{
   "scenario":{
      "desc":"Training Classification travel data (small)",
      "type":"classification",
      "language":"en",
      "business_object":"ticket"
   },
   "tabulating":[
      {
         "source":"description",
         "topic":"category"
      }
   ],
   "options":{
      "training_file_field_delimiter":";"
   },
   "mapping":{
      "input":[
         "description"
      ],
      "output":[
         "category"
      ]
   },
   "training":{
      "file":"base64decoded data making use of the delimiter specified in
training_file_field_delimiter"
   }
}
```

6.1.3.6 Data Extractor for Classification Type Training

Example

Sample Request (with Transformations Object)

```
{
   "scenario":{
      "desc":"Text Classification",
      "type":"classification",
      "language":"en",
      "business_object":"ticket"
   },
   "mapping":{
      "input":[
         "Name",
         "Text"
      ],
      "output":[
         "ProductID","ProductCategoryDescription","ReporterPartyID","AssignedTo"
      ]
   },
  "transformations":[
  {
    "output_category": "ProductID",
    "operation": {
      "operation_type": "replace",
      "rules": [
        {
          "category_class": "CA_21",
          "new_category_class": "CA_3"
        },
        {
          "category_class": "CO-ACC",
          "new_category_class": "CA_3"
        }
      ]
    }
  },
   {
    "output_category": "ProductCategoryDescription",
    "operation": {
      "operation_type": "replace",
      "rules": [
        {
          "category_class": "CA_21"
        },
        {
          "category_class": "CO-ACC"
        }
      ]
    }
    },
    {
    "output_category": "ReporterPartyID",
    "operation": {
      "operation_type": "replace",
      "rules": [
        {
          "category_class": "CA_21"
        },
```

```
        {
          "category_class": "CO-ACC"
        }
      ]
    }
  },
  {
    "output_category": "AssignedTo",
    "operation": {
      "operation_type": "replace",
      "rules": [
        {
          "category_class": "8000000013"
        },
        {
          "category_class": "XXXX-YYY-ZZZ"
        }
      ]
    }
  }
  ],
   "odata":{
 "credential":
        {
         "service_url":"https://xxx/xxxOdata",
            "authorization":"Basic XXXXX="
     },
      "application":"C4C",
      "query_options":{
         "entity":"ServiceRequestCollection",

"select":"Name,ProductID,ProductCategoryDescription,ReporterPartyID,AssignedTo",
         "filter":"ETag ge datetimeoffset'2018-05-09T01:00:00.00Z' and ETag le
datetimeoffset'2018-05-10T17:00:00.00Z'",
         "orderby":"",
         "expand":"",
         "batch_query_options":{
            "entity":"ServiceRequestDescription",
            "select":"Text",
            "filter":"TypeCode eq '10004'",
            "orderby":"",
            "expand":""
         }
      }
   }
}
```

Sample Response

```
{
    "business_object": "ticket",
    "desc": "Text Classification",
    "file_upload_timestamp": "2018-10-04 04:39:22 UTC",
    "language": "en",
    "model_id": "8477ba35308840ab8dbfa5ad4a4f2e83",
    "model_status": "DATA_EXTRACTION_PENDING",
    "status": 0,
    "status_message": "ok",
    "type": "classification"
}
```

Data Extractor Sample Request (with template)

```
{
   "scenario": {
      "name": "Training Initial data",
```

```
      "type": "classification",
      "language": "en",
      "business_object": "ticket"
   },
   "mapping": {
      "input": [
         "Name",
         "Text"
      ],
      "output": [
         "ServiceIssueCategoryID"
      ]
   },
   "odata": {
      "credential": {
         "service_url": "https://my******.crm.ondemand.com/sap/c4c/odata/v1/
c4codata",
         "authorization": "Basic ************************************"
      },
      "application": "C4C",
      "template": "ServiceRequest",
      "date_from": "2017-01-01T04:00:00.00Z"
   }
}
```

Example

Data Extractor Sample Request (without template)

```
{
    "scenario":{
        "name":"TrainingInitialdata",
        "type":"classification",
        "language":"en",
        "business_object":"ticket"
    },
    "mapping":{
        "input":[
            "Name",
            "Text"
        ],
        "output":[
            "ServiceIssueCategoryID"
        ]
    },
    "odata":{
        "credential":{
                  "service_url":"https://my******.crm.ondemand.com/sap/c4c/
odata/v1/c4codata",
    "authorization":"Basic ************************************"
    },
    "application":"C4C",
    "query_options":{
            "entity":"ServiceRequestCollection",
            "select":"Name,ServiceIssueCategoryID",
            "filter":"ServiceIssueCategoryID ne null and ETag ge
datetimeoffset'2017-01-01T04:00:00.00Z'",
            "orderby":"",
            "expand":"",
            "batch_query_options":{
                "entity":"ServiceRequestDescription",
                "select":"Text",
                "filter":"TypeCode eq '10004'",
```

```
                "orderby":"",
                "expand":""
            }
        }
    }
}
```

Example

Data Extractor Sample Response

Positive Response

```
{
    "status_message": "ok",
    "model_id": "5954b0448e18510007f9ecac",
    "status": 0
}
```

Negative Response (odata info should not be empty)

```
{
  "status": 43,
  "status_message": "odata info should not be empty"
}
```

Negative Response (odata credential should not be empty)

```
{
  "status": 44,
  "status_message": "odata credential should not be empty"
}
```

Negative Response (odata service url should not be empty)

```
{
  "status": 45,
  "status_message": "odata service URL should not be empty"
}
```

Negative Response (odata authorization info should not be empty)

```
{
  "status": 46,
  "status_message": "odata authorization info should not be empty"
}
```

Negative Response (odata application info is should not be empty)

```
{
  "status": 47,
  "status_message": "odata application info is should not be empty"
}
```

Negative Response (invalid application)

```
{
  "status": 48,
```

```
  "status_message": "invalid application"
}
```

Negative Response (invalid odata template type)

```
{
  "status": 49,
  "status_message": "invalid odata template type"
}
```

Negative Response (invalid date format)

```
{
  "status": 50,
  "status_message": "invalid date format"
}
```

Negative Response (either *date_from* **or** *date_to* **is allowed)**

```
{
  "status": 53,
  "status_message": "either date_from or date_to is allowed"
}
```

Negative Response (odata query options should not be empty or provide template)

```
{
  "status": 54,
  "status_message": "odata query options should not be empty or provide template"
}
```

Negative Response (odata entity name should not be empty)

```
{
  "status": 55,
  "status_message": "odata entity name should not be empty"
}
```

Negative Response (odata select option should not be empty)

```
{
  "status": 56,
  "status_message": "odata select option should not be empty"
}
```

Negative Response (input/output mapping is not same as select option)

```
{
  "status": 57,
  "status_message": "input/output mapping is not same as select option"
}
```

Negative Response (data extractor call failed without response)

```
{
  "status": 59,
  "status_message": "data extractor call failed without response"
}
```

6.1.3.7 Upload File with Destination Configuration for OData Extraction

Example

Sample Request

```
{
   "scenario":{
      "desc":"Text Classification",
      "type":"classification",
      "language":"en",
      "business_object":"ticket"
   },
   "mapping":{
      "input":[
         "Name",
         "Text"
      ],
      "output":[
         "ProductID","ProductCategoryDescription","ReporterPartyID","AssignedTo"
      ]
   },
 "odata":{
 "credential":{
                "service_url": "x-sap-destination: sample"
      },
      "application":"C4C",
      "query_options":{
         "entity":"ServiceRequestCollection",

"select":"Name,ProductID,ProductCategoryDescription,ReporterPartyID,AssignedTo",
         "filter":"ETag ge datetimeoffset'2019-12-09T01:00:00.00Z' and ETag le
datetimeoffset'2020-01-11T17:00:00.00Z'",
         "orderby":"",
         "expand":"",
         "batch_query_options":{
            "entity":"ServiceRequestDescription",
            "select":"Text",
            "filter":"TypeCode eq '10004'",
            "orderby":"",
            "expand":""
         }
      }
   },
 }
```

6.1.3.8 Data Extractor with Priority for Priority Type Training

Example

Sample Request for Priority Type Training with Priority

```
{
   "scenario":{
      "desc":"Text Classification",
      "type":"classification",
      "language":"en",
      "business_object":"ticket"
   },
   "mapping":{
      "input":[
         "Name",
         "Text"
      ],
      "output":[
         "ObjectID"
      ],

    "priority":"ObjectID",
   },
   "odata":{
                "credential":{
                     "service_url":"https://example.com",
                     "authorization":"Basic QWERtWRtaW5pc3RyYXRpb24wM=="
                  },
      "application":"XXX",
      "query_options":{
         "entity":"ServiceRequestCollection",
         "select":"Name,ObjectID",
         "filter":"ETag ge datetimeoffset'2018-09-07T07:32:10.00Z' and ETag le
datetimeoffset'2018-09-07T08:53:35.00Z'",
         "orderby":"",
         "expand":"",
         "batch_query_options":{
            "entity":"ServiceRequestDescription",
            "select":"Text",
            "filter":"TypeCode eq '10004'",
            "orderby":"",
            "expand":""
         }
      }
   }
}
```

6.1.3.9 Data Extractor with Priority and Keywords (Relabeling) for Priority Type Training

When there is, for example, a large percentage of "Normal" priority labels compared to "Urgent" priority, using the priority and keywords (relabeling) may aid in improving the confidence score of the "Urgent" priority

prediction. This involves performing an OData extraction request with priority and keywords (relabeling of data according to certain keywords). Consider the following when there is a distributional imbalance of the priority labeled dataset:

- The columns indicate where to look for the search text in a case-insensitive manner. The text might occur in any of the columns (in case more than one is provided) for relabeling to happen.
- If multiple texts are provided and any of the provided ones are in either of the input columns then relabeling will also take place.
- If you add `keywords` in the OData request body, the `priority` field becomes mandatory for relabeling to work.

In the example below, there are three sections in the `keywords` mapping object attribute and each one indicates the following:

- Either of Name or Text contains “How can” or “Windows” irrespective of case, relabel the ObjectID (the priority label in this case) to ”Urgent”.
- Either of Name or Text contains “How do” or “desktop” irrespective of case, relabel the ObjectID (the priority label in this case) to “Not Urgent”.
- Either of Name or Text contains “Windows” or “Error” or ”Overnight”, relabel the ObjectID (the priority label in this case) to “Immediate”.

Example

Sample Request for Priority Type Training with Priority and Keywords

```
{
   "scenario":{
      "desc":"Text Classification",
      "type":"classification",
      "language":"en",
      "business_object":"ticket"
   },
   "mapping":{
      "input":[
         "Name",
         "Text"
      ],
      "output":[
         "ObjectID"
      ],

    "priority":"ObjectID",
    "keywords":[
        {
            "columns":[
                "Name",
                "Text"
            ],
            "text":[
                "How can",
                "Windows"
            ],
            "value":"Urgent"
        },
        {
            "columns":[
            "Name",
```

```
            "Text"
            ],
            "text":[
            "How do",
            "desktop"
            ],
            "value":"Not Urgent"
        },
        {
            "columns":[
            "Name",
            "Text"
            ],
            "text":[
            "Windows",
            "Error",
            "Overnight",
            ],
            "value":"Immediate"
        }

    ]
   },
   "odata":{
     "credential":{
                        "service_url":"https://example.com/sap/c4c/odata/v1/
c4codata",
                        "authorization":"Basic QWRtaWsomeauthorization=="
                    },
      "application":"some_app",
      "query_options":{
         "entity":"Request_Parent",
         "select":"Name,ObjectID",
         "filter":"ETag ge datetimeoffset'2018-09-07T07:32:10.00Z' and ETag le
datetimeoffset'2018-09-07T08:53:35.00Z'",
         "orderby":"",
         "expand":"",
         "batch_query_options":{
            "entity":"Request_Child",
            "select":"Text",
            "filter":"TypeCode eq '10004'",
            "orderby":"",
            "expand":""
         }
      }
   }
}
```


6.1.3.10 Upload File with Groupby for Classification Type Training

Example


```
{
   "scenario":{
      "desc":"Training Tui like data",
```

```
      "type":"classification",
      "language":"en",
      "business_object":"ticket"
   },
   "mapping":{
      "input":[
         "Product",
         "Consumer_complaint_narrative"
      ],
      "output":[
         "Sub-product"
      ],
      "groupby":"Product"
   },
   "training":{
      "file":"base64decoded data"
   }
}
```

6.1.3.11 Upload File with Encoding Option for Classification Type Training

Example

Sample Request with Latin-1 encodings

```
{
   "scenario":{
      "desc":"Training Classification travel data (small)",
      "type":"classification",
      "language":"en",
      "business_object":"ticket"
   },
   "tabulating":[
      {
         "source":"description",
         "topic":"category"
      }
   ],
   "options":{
      "encoding":"latin-1"
   },
   "mapping":{
      "input":[
         "description"
      ],
      "output":[
         "category"
      ]
   },
   "training":{
      "file":"latin-1 encoded file in base64"
   }
}
```

6.1.3.12 Upload File for Clustering Type Training

Example

Sample Request

```
{
   "scenario":{
      "name":"Training clustering model",
      "type":"clustering",
      "language":"en",
      "business_object":"ticket",
      "desc":"Clustering file upload"
   },
   "mapping":{
      "input":[
         "title",
         "content"
      ],
      "output":[
         "id",
         "title",
         "label",
         "timestamp"
      ]
   },
   "training":{
      "file":"<<base64EncodedData>>"
   }
}
```

Example

Sample Response

```
{
   "business_object":"ticket",
   "desc":"Clustering file upload",
   "disabled_category_class":[

   ],
   "enabled_category_class":[

   ],
   "extracted_input":[
      {
         "field":"title",
         "type":"str"
      },
      {
         "field":"content",
         "type":"str"
      }
   ],
   "extracted_output":[
```

```
      {
         "field":"id"
      },
      {
         "field":"title"
      },
      {
         "field":"label"
      },
      {
         "field":"timestamp"
      }
   ],
   "file_upload_timestamp":"2021-02-15 08:18:24 UTC",
   "incremental":false,
   "incremental_possible":false,
   "language":"en",
   "model_id":"6ea3d6a6cddf45bc84577715de722bdf",
   "model_status":"NEW",
   "new_category_class":[

   ],
   "records_count":2001,
   "status":0,
   "status_message":"ok",
   "tabulations_count":0,
   "type":"clustering"
}
```


6.1.3.13 Bulk File Upload

Example

Sample Request

```
{
   "scenario":{
      "desc":"Training Classification travel data with priority and keywords
(small)",
      "type":"classification",
      "language":"en",
      "business_object":"ticket"
   },
   "options":{
      "bulk_file_upload":true,
      "bulk_file_upload_complete":false
   },
   "mapping":{
      "input":[
         "description"
      ],
      "output":[
         "category"
      ],
      "keywords":[
         {
            "columns":[
```

```
                "description"
            ],
            "text":[
                "Name",
                "Surname"
            ],
            "value":"1"
        }
    ]
  },
  "training":{
    "file":"
c3ViamVjdCxib2R5LGFydGljbGVfaWQsYXJ0aWNsZV9kZXNjLGRhdGFzb3VyY2UsYXJ0aWNsZV91cmwKI
kknbSBhIHRyaXBsZSBDYXByaWNvcm4gKFN1biwgTW9vbiBhbmQgYXNjZW5kYW50IGluIENhcHJpY29ybi
kgV2hhdCBkb2VzIHRoaXMgc2F5IGFib3V0IG1lPyIsLDEscmVsaWdpb3VzIHNlbnRpbWVudHMsbWluZHR
vdWNoLHd3dy50aW1lc29maW5kaWEuY29tCldoYXQgc2hvdWxkIEkgZG8gdG8gYmUgYSBncmVhdCBnZW9s
b2dpc3Q"
  }
}
```


Example

Sample Response


```
{
  "business_object":"ticket",
  "desc":"Training Classification travel data with priority and keywords
(small)",
  "file_upload_timestamp":"2021-05-06 17:25:18 UTC",
  "incremental":false,
  "incremental_possible":false,
  "language":"en",
  "model_id":"90c6daa1894143dda96410bd1e1b70c7",
  "model_status":"DATA_UPLOAD_IN_PROGRESS",
  "options":{
    "bulk_file_upload":{
      "complete":false,
      "file_count":1
    }
  },
  "status":0,
  "status_message":"ok",
  "type":"classification"
}
```


Example

Subsequent Payload Sample Request (with `bulk_file_upload_complete` set to false)


```
{
  "bulk_file_upload_reference_model_id":"90c6daa1894143dda96410bd1e1b70c7",
  "bulk_file_upload_complete":false,
  "training":{
    "file":
“c3ViamVjdCxib2R5LGFydGljbGVfaWQsYXJ0aWNsZV9kZXNjLGRhdGFzb3VyY2UsYXJ0aWNsZV91cmwK
IkknbSBhIHRyaXBsZSBDYXByaWNvcm4gKFN1biwgTW9vbiBhbmQgYXNjZW5kYW50IGluIENhcHJpY29yb
ikgV2hhdCBkb2VzIHRoaXMgc2F5IGFib3V0IG1lPyIsLDEscmVsaWdpb3VzIHNlbnRpbWVudHMsbWluZH
```

```
RvdWNoLHd3dy50aW1lc29maW5kaWEuY29tCldoYXQgc2hvdWxkIEkgZG8gdG8gYmUgYSBncmVhdCBnZW9
sb2dpc3Q”
   }
}
```


Example

Subsequent Payload Sample Response (with `bulk_file_upload_complete` set to false)


```
{
   "business_object":"ticket",
   "desc":"Training Classification travel data with priority and keywords
(small)",
   "file_upload_timestamp":"2021-05-06 17:25:18 UTC",
   "incremental":false,
   "incremental_possible":false,
   "language":"en",
   "model_id":"90c6daa1894143dda96410bd1e1b70c7",
   "model_status":"DATA_UPLOAD_IN_PROGRESS",
   "options":{
      "bulk_file_upload":{
         "complete":false,
         "file_count":2
      }
   },
   "status":0,
   "status_message":"ok",
   "type":"classification"
}
```


Example

Subsequent Payload Sample Request (with `bulk_file_upload_complete` set to true)


```
{
   "bulk_file_upload_reference_model_id":"90c6daa1894143dda96410bd1e1b70c7",
   "bulk_file_upload_complete":true,
   "training":{
      "file":
“c3ViamVjdCxib2R5LGFydGljbGVfaWQsYXJ0aWNsZV9kZXNjLGRhdGFzb3VyY2UsYXJ0aWNsZV91cmwK
IkknbSBhIHRyaXBsZSBDYXByaWNvcm4gKFN1biwgTW9vbiBhbmQgYXNjZW5kYW50IGluIENhcHJpY29yb
ikgV2hhdCBkb2VzIHRoaXMgc2F5IGFib3V0IG1lPyIsLDEscmVsaWdpb3VzIHNlbnRpbWVudHMsbWluZH
RvdWNoLHd3dy50aW1lc29maW5kaWEuY29tCldoYXQgc2hvdWxkIEkgZG8gdG8gYmUgYSBncmVhdCBnZW9
sb2dpc3Q”
   }
}
```


Example

Subsequent Payload Sample Response (with `bulk_file_upload_complete` set to true)


```
{
```

```
    "business_object": "ticket",
    "desc": "Training Classification travel data with priority and keywords
(small)",
    "file_upload_timestamp": "2021-05-06 17:25:18 UTC",
    "language": "en",
    "model_id": "90c6daa1894143dda96410bd1e1b70c7",
    "model_status": "DATA_PREPROCESSING_PENDING",
    "status": 0,
    "status_message": "ok",
    "type": "classification"
}
```


Example

Subsequent Payload Sample Response (from Get Model Status once preprocessing is complete)


```
{
   "business_object":"ticket",
   "combined_accuracy":0.0,
   "data_preprocessing_end_timestamp":"2021-05-06 17:34:38 UTC",
   "data_preprocessing_start_timestamp":"2021-05-06 17:34:37 UTC",
   "desc":"Training Classification travel data with priority and keywords
(small)",
   "disabled_category_class":[
      {
         "category_class":[

         ],
         "field":"category"
      }
   ],
   "enabled_category_class":[
      {
         "category_class":[
            "complaint",
            "request",
            "compliment"
         ],
         "field":"category"
      }
   ],
   "extracted_input":[
      {
         "field":"description",
         "type":"str"
      }
   ],
   "extracted_output":[
      {
         "distribution":[
            {
               "count":105,
               "value":"complaint"
            },
            {
               "count":108,
               "value":"request"
            },
            {
               "count":78,
               "value":"compliment"
            }
         ],
```

```
            "field":"category"
         }
      ],
      "file_upload_timestamp":"2021-05-06 17:25:18 UTC",
      "incremental":false,
      "incremental_possible":true,
      "language":"en",
      "model_id":"90c6daa1894143dda96410bd1e1b70c7",
      "model_status":"NEW",
      "new_category_class":[
         {
            "category_class":[

            ],
            "field":"category"
         }
      ],
      "options":{
         "bulk_file_upload":{
            "complete":true,
            "file_count":3
         }
      },
      "records_count":291,
      "status":0,
      "status_message":"ok",
      "tabulations_count":0,
      "type":"classification"
   }
```


6.2 Get Model Status

Returns the training results (model details) using the model identifier obtained in List All Models [page 88] or Upload Training File / Data Extraction [page 20]. See Get Model Accuracy [page 96] for model training accuracy or confusion matrix.

Key	Value
HTTP Method	GET
Path	model/status
Headers	Authorization: Bearer <token> X-Request-ID: <unique string>

Related Information

Input Parameters [page 67]
Expected Response [page 67]
Examples [page 67]

6.2.1 Input Parameters

The parameters are given as URL parameters in the request with the following entries:

Parameter	Type	Description
`model_id`	String	Unique identifier for the model.

6.2.2 Expected Response

The call returns a model status object (similar to Upload Training File / Data Extraction [page 20]) showing the model status and training progress. See Model Status Object [page 37].

> **i Note**
>
> The same fields will be returned in the response body of Upload Training File / Data Extraction [page 20] and Get Model Status [page 66].

6.2.3 Examples

Example

Sample Request

```
GET /sti/training/model/status?model_id=5836718c1925eb000dc6810e HTTP/1.1
Authorization: Bearer ed195a25c62942aa85dce785d8d33496
X-Request-ID: 440622E1-E0F4-4A60-A1A7-172090AA739A.2147483648.2147516416
```

Example

Sample Responses

- New Uploaded Training File, neither Trained nor Activated [page 68]
- Model Training Currently in Progress [page 70]
- Training is Complete and the Model is READY for Activation [page 72]
- Trained Uploaded File and Model is LOADING [page 74]

- Trained Uploaded File and Model is ACTIVE [page 76]
- New Uploaded Incremental Training File, neither Trained nor Activated [page 78]
- Incremental Training is Complete and the Model is READY for Activation [page 80]
- Get Model Status for Recommendation with Language Mix [page 82]

6.2.3.1 New Uploaded Training File, neither Trained nor Activated

Example

Sample Response

New uploaded training file, neither trained nor activated:


```
{
  "model_id": "5836718c1925eb000dc6810e",
  "model_status": "NEW",
  "desc": "Simple Twitter Training Data",
  "type": "classification",
  "business_object": "ticket",
  "language": "en",
  "tabulations_count": 2,
  "records_count": 19794,
  "incremental": false,
  "file_upload_timestamp": "2017-05-17 04:56:30 UTC",
  "extracted_input": [
    {
      "type": "str",
      "field": "subject"
    },
    {
      "type": "str",
      "field": "body"
    }
  ],
  "extracted_output": [
    {
      "field": "ticket_type",
      "distribution": [
        {
          "value": "complaint",
          "count": 7132
        },
        {
          "value": "request",
          "count": 6270
        },
        {
          "value": "compliment",
          "count": 6392
        }
      ]
    },
    {
      "field": "routing_inbox",
```

```
      "distribution": [
        {
          "value": "reservations",
          "count": 3412
        },
        {
          "value": "product_support",
          "count": 6123
        },
        {
          "value": "housekeeping",
          "count": 5141
        },
        {
          "value": "public_relations",
          "count": 5118
        }
      ]
    }
  ],
  "new_category_class": [
   {
     "field": "ticket_type",
     "category_class": []
   },
   {
     "field": "routing_inbox",
     "category_class": []
   }
  ],
  "enabled_category_class": [
   {
     "field": "ticket_type",
     "category_class": [
       "complaint",
       "request",
       "compliment"
     ]
   },
   {
     "field": "routing_inbox",
     "category_class": [
      "reservations",
      "product_support",
      "housekeeping",
      "public_relations"
      ]
    }
  ],
  "disabled_category_class": [
   {
     "field": "ticket_type",
     "category_class": []
   },
   {
     "field": "routing_inbox",
     "category_class": []
   }
  ],
  "status": 0,
  "status_message": "ok"
}
```

6.2.3.2 Model Training Currently in Progress

Example

Sample Response

Model training currently in progress:

```
{
  "model_id": "5836718c1925eb000dc6810e",
  "model_status": "IN_TRAINING",
  "desc": "Simple Twitter Training Data",
  "type": "classification",
  "business_object": "ticket",
  "language": "en",
  "tabulations_count": 2,
  "records_count": 19794,
  "file_upload_timestamp": "2017-05-17 04:56:30 UTC",
  "training_start_timestamp": "2017-05-17 05:01:30 UTC",
  "incremental": false,
  "extracted_input": [
    {
      "type": "str",
      "field": "subject"
    },
    {
      "type": "str",
      "field": "body"
    }
  ],
  "extracted_output": [
    {
      "field": "ticket_type",
      "distribution": [
        {
          "value": "complaint",
          "count": 7132
        },
        {
          "value": "request",
          "count": 6270
        },
        {
          "value": "compliment",
          "count": 6392
        }
      ]
    },
    {
      "field": "routing_inbox",
      "distribution": [
        {
          "value": "reservations",
          "count": 3412
        },
        {
          "value": "product_support",
          "count": 6123
        },
        {
          "value": "housekeeping",
```

```
          "count": 5141
        },
        {
          "value": "public_relations",
          "count": 5118
        }
      ]
    }
  ],
  "new_category_class": [
   {
     "field": "ticket_type",
     "category_class": []
   },
   {
     "field": "routing_inbox",
     "category_class": []
   }
  ],
  "enabled_category_class": [
   {
     "field": "ticket_type",
     "category_class": [
       "complaint",
       "request",
       "compliment"
     ]
   },
   {
     "field": "routing_inbox",
     "category_class": [
      "reservations",
      "product_support",
      "housekeeping",
      "public_relations"
      ]
    }
  ],
  "disabled_category_class": [
   {
     "field": "ticket_type",
     "category_class": []
   },
   {
     "field": "routing_inbox",
     "category_class": []
   }
  ],
  "progress_level": 12,
  "status": 0,
  "status_message": "ok"
}
```

6.2.3.3 Training is Complete and the Model is READY for Activation

Example

Sample Response

Training is complete and the model is READY for activation:


```
{
  "model_id": "5836718c1925eb000dc6810e",
  "model_status": "READY",
  "desc": "Simple Twitter Training Data",
  "type": "classification",
  "business_object": "ticket",
  "language": "en",
  "tabulations_count": 2,
  "records_count": 19794,
  "incremental": false,
  "incremental_possible": true,
  "file_upload_timestamp": "2017-05-17 04:56:30 UTC",
  "training_start_timestamp": "2017-05-17 05:01:30 UTC",
  "training_finish_timestamp": "2017-05-17 05:12:42 UTC",
  "extracted_input": [
    {
      "type": "str",
      "field": "subject"
    },
    {
      "type": "str",
      "field": "body"
    }
  ],
  "extracted_output": [
    {
      "field": "ticket_type",
      "distribution": [
        {
          "value": "complaint",
          "count": 7132
        },
        {
          "value": "request",
          "count": 6270
        },
        {
          "value": "compliment",
          "count": 6392
        }
      ]
    },
    {
      "field": "routing_inbox",
      "distribution": [
        {
          "value": "reservations",
          "count": 3412
        },
        {
          "value": "product_support",
```

```
          "count": 6123
        },
        {
          "value": "housekeeping",
          "count": 5141
        },
        {
          "value": "public_relations",
          "count": 5118
        }
      ]
    }
  ],
  "new_category_class": [
   {
     "field": "ticket_type",
     "category_class": []
   },
   {
     "field": "routing_inbox",
     "category_class": []
   }
  ],
  "enabled_category_class": [
   {
     "field": "ticket_type",
     "category_class": [
       "complaint",
       "request",
       "compliment"
     ]
   },
   {
     "field": "routing_inbox",
     "category_class": [
      "reservations",
      "product_support",
      "housekeeping",
      "public_relations"
      ]
    }
  ],
  "disabled_category_class": [
   {
     "field": "ticket_type",
     "category_class": []
   },
   {
     "field": "routing_inbox",
     "category_class": []
   }
  ],
  "combined_accuracy": 0.8923916338284329,
  "status": 0,
  "status_message": "ok"
}
```

6.2.3.4 Trained Uploaded File and Model is LOADING

Example

Sample Response

Trained uploaded file and model is LOADING:


```
{
  "model_id": "5836718c1925eb000dc6810e",
  "model_status": "LOADING",
  "desc": "Simple Twitter Training Data",
  "type": "classification",
  "business_object": "ticket",
  "language": "en",
  "tabulations_count": 2,
  "records_count": 19794,
  "incremental": false,
  "incremental_possible": true,
  "file_upload_timestamp": "2017-05-17 04:56:30 UTC",
  "training_start_timestamp": "2017-05-17 05:01:30 UTC",
  "training_finish_timestamp": "2017-05-17 05:12:42 UTC",
  "model_activation_timestamp": "2017-05-17 05:25:15 UTC",
  "extracted_input": [
    {
      "type": "str",
      "field": "subject"
    },
    {
      "type": "str",
      "field": "body"
    }
  ],
  "extracted_output": [
    {
      "field": "ticket_type",
      "distribution": [
        {
          "value": "complaint",
          "count": 7132
        },
        {
          "value": "request",
          "count": 6270
        },
        {
          "value": "compliment",
          "count": 6392
        }
      ]
    },
    {
      "field": "routing_inbox",
      "distribution": [
        {
          "value": "reservations",
          "count": 3412
        },
        {
          "value": "product_support",
          "count": 6123
```

```
        },
        {
          "value": "housekeeping",
          "count": 5141
        },
        {
          "value": "public_relations",
          "count": 5118
        }
      ]
    }
  ],
  "new_category_class": [
   {
     "field": "ticket_type",
     "category_class": []
   },
   {
     "field": "routing_inbox",
     "category_class": []
   }
  ],
  "enabled_category_class": [
   {
     "field": "ticket_type",
     "category_class": [
       "complaint",
       "request",
       "compliment"
     ]
   },
   {
     "field": "routing_inbox",
     "category_class": [
      "reservations",
      "product_support",
      "housekeeping",
      "public_relations"
      ]
    }
  ],
  "disabled_category_class": [
   {
     "field": "ticket_type",
     "category_class": []
   },
   {
     "field": "routing_inbox",
     "category_class": []
   }
  ],
  "combined_accuracy": 0.8923916338284329,
  "status": 0,
  "status_message": "ok"
}
```

6.2.3.5 Trained Uploaded File and Model is ACTIVE

Example

Sample Response

Trained uploaded file and model is ACTIVE:


```
{
  "model_id": "5836718c1925eb000dc6810e",
  "model_status": "ACTIVE",
  "desc": "Simple Twitter Training Data",
  "type": "classification",
  "business_object": "ticket",
  "language": "en",
  "tabulations_count": 2,
  "records_count": 19794,
  "incremental": false,
  "incremental_possible": true,
  "file_upload_timestamp": "2017-05-17 04:56:30 UTC",
  "training_start_timestamp": "2017-05-17 05:01:30 UTC",
  "training_finish_timestamp": "2017-05-17 05:12:42 UTC",
  "model_activation_timestamp": "2017-05-17 05:25:15 UTC",
  "extracted_input": [
    {
      "type": "str",
      "field": "subject"
    },
    {
      "type": "str",
      "field": "body"
    }
  ],
  "extracted_output": [
    {
      "field": "ticket_type",
      "distribution": [
        {
          "value": "complaint",
          "count": 7132
        },
        {
          "value": "request",
          "count": 6270
        },
        {
          "value": "compliment",
          "count": 6392
        }
      ]
    },
    {
      "field": "routing_inbox",
      "distribution": [
        {
          "value": "reservations",
          "count": 3412
        },
        {
          "value": "product_support",
          "count": 6123
```

```
        },
        {
          "value": "housekeeping",
          "count": 5141
        },
        {
          "value": "public_relations",
          "count": 5118
        }
      ]
    }
  ],
  "new_category_class": [
   {
     "field": "ticket_type",
     "category_class": []
   },
   {
     "field": "routing_inbox",
     "category_class": []
   }
  ],
  "enabled_category_class": [
   {
     "field": "ticket_type",
     "category_class": [
       "complaint",
       "request",
       "compliment"
     ]
   },
   {
     "field": "routing_inbox",
     "category_class": [
      "reservations",
      "product_support",
      "housekeeping",
      "public_relations"
      ]
    }
  ],
  "disabled_category_class": [
   {
     "field": "ticket_type",
     "category_class": []
   },
   {
     "field": "routing_inbox",
     "category_class": []
   }
  ],
  "combined_accuracy": 0.8923916338284329,
  "status": 0,
  "status_message": "ok"
}
```

6.2.3.6 New Uploaded Incremental Training File, neither Trained nor Activated

Example

Sample Response

New uploaded incremental training file, neither trained nor activated:


```
{
  "model_id": "5836718c1925eb0110dc6810e",
  "model_status": "NEW",
  "desc": "Simple Twitter Incremental Training Data",
  "type": "classification",
  "business_object": "ticket",
  "language": "en",
  "tabulations_count": 2,
  "records_count": 39588,
  "incremental": true,
  "incremental_upload_timestamp": "2017-05-17 04:56:30 UTC",
  "extracted_input": [
    {
      "type": "str",
      "field": "subject"
    },
    {
      "type": "str",
      "field": "body"
    }
  ],
  "extracted_output": [
    {
      "field": "ticket_type",
      "distribution": [
        {
          "value": "complaint",
          "count": 14264
        },
        {
          "value": "request",
          "count": 6270
        },
        {
          "value": "compliment",
          "count": 12784
        },
        {
          "value": "query",
          "count": 6270
        }
      ]
    },
    {
      "field": "routing_inbox",
      "distribution": [
        {
          "value": "reservations",
          "count": 3412
        },
        {
```

```
          "value": "product_support",
          "count": 6123
        },
        {
          "value": "housekeeping",
          "count": 5141
        },
        {
          "value": "public_relations",
          "count": 5118
        }
      ]
    }
  ],
  "new_category_class": [
   {
     "field": "ticket_type",
     "category_class": ["query"]
   },
   {
     "field": "routing_inbox",
     "category_class": []
   }
  ],
  "enabled_category_class": [
   {
     "field": "ticket_type",
     "category_class": [
       "complaint",
       "request",
       "query",
       "compliment"
     ]
   },
   {
     "field": "routing_inbox",
     "category_class": [
      "reservations",
      "product_support",
      "housekeeping",
      "public_relations"
      ]
    }
  ],
  "disabled_category_class": [
   {
     "field": "ticket_type",
     "category_class": []
   },
   {
     "field": "routing_inbox",
     "category_class": []
   }
  ],
  "status": 0,
  "status_message": "ok"
}
```

6.2.3.7 Incremental Training is Complete and the Model is READY for Activation

Example

Sample Response

Incremental training is complete and the model is READY for activation:


```
{
  "model_id": "5836718c1925eb0110dc6810e",
  "model_status": "READY",
  "desc": "Simple Twitter Incremental Training Data",
  "type": "classification",
  "business_object": "ticket",
  "language": "en",
  "tabulations_count": 2,
  "records_count": 39588,
  "incremental": true,
  "incremental_possible": true,
  "incremental_upload_timestamp": "2017-05-17 04:56:30 UTC",
  "training_start_timestamp": "2017-05-17 05:01:30 UTC",
  "training_finish_timestamp": "2017-05-17 05:12:42 UTC",
  "extracted_input": [
    {
      "type": "str",
      "field": "subject"
    },
    {
      "type": "str",
      "field": "body"
    }
  ],
  "extracted_output": [
    {
      "field": "ticket_type",
      "distribution": [
        {
          "value": "complaint",
          "count": 14264
        },
        {
          "value": "request",
          "count": 6270
        },
        {
          "value": "compliment",
          "count": 12784
        },
        {
          "value": "query",
          "count": 6270
        }
      ]
    },
    {
      "field": "routing_inbox",
      "distribution": [
        {
          "value": "reservations",
```

```
          "count": 3412
        },
        {
          "value": "product_support",
          "count": 6123
        },
        {
          "value": "housekeeping",
          "count": 5141
        },
        {
          "value": "public_relations",
          "count": 5118
        }
      ]
    }
  ],
  "new_category_class": [
   {
     "field": "ticket_type",
     "category_class": []
   },
   {
     "field": "routing_inbox",
     "category_class": []
   }
  ],
  "enabled_category_class": [
   {
     "field": "ticket_type",
     "category_class": [
       "complaint",
       "request",
       "query",
       "compliment"
     ]
   },
   {
     "field": "routing_inbox",
     "category_class": [
      "reservations",
      "product_support",
      "housekeeping",
      "public_relations"
      ]
    }
  ],
  "disabled_category_class": [
   {
     "field": "ticket_type",
     "category_class": []
   },
   {
     "field": "routing_inbox",
     "category_class": []
   }
  ],
  "status": 0,
  "status_message": "ok"
}
```

6.2.3.8 Get Model Status for Recommendation with Language Mix

Example

Sample Response

Get model status for recommendation with language mix:


```
{
    "Recall": [
        {
            "language": "en",
            "recall_rate": 100
        },
        {
            "language": "de",
            "recall_rate": 100
        },
        {
            "language": "es",
            "recall_rate": 90
        }
    ],
    "Recall@10": 100,
    "business_object": "ticket",
    "desc": "Training Recommendation(multi language)",
    "disabled_category_class": [],
    "enabled_category_class": [],
    "extracted_input": [
        {
            "field": "subject",
            "type": "str"
        }
    ],
    "extracted_output": [
        {
            "field": "description"
        },
        {
            "field": "article_id"
        },
        {
            "field": "article_desc"
        },
        {
            "field": "datasource"
        },
        {
            "field": "article_url"
        }
    ],
    "file_upload_timestamp": "2019-08-23 09:25:43 UTC",
    "incremental": false,
    "incremental_possible": true,
    "language": "en",
    "language_mix": [
        "en",
        "de",
        "es"
```

```
    ],
    "model_activation_timestamp": "2019-08-28 03:32:55 UTC",
    "model_deactivate_timestamp": "2019-08-28 04:19:45 UTC",
    "model_id": "8ed7cb4e354a47069795d3c3edd65b24",
    "model_status": "READY",
    "new_category_class": [],
    "records_count": 300,
    "status": 0,
    "status_message": "ok",
    "tabulations_count": 0,
    "training_finish_timestamp": "2019-08-23 10:04:41 UTC",
    "training_start_timestamp": "2019-08-23 10:04:37 UTC",
    "training_submit_timestamp": "2019-08-23 10:03:51 UTC",
    "type": "recommendation"
```


6.3 Start Model Training

Initiates the training process for a given uploaded training file. Model training is a background process, so it may not complete immediately after the call returns. Upon return, the model status should change to IN_TRAINING. When the training process completes, the model status should change to either READY or INVALID. INVALID status would be the case when the training file does not contain sufficient data to construct a machine-learning model. Upon training process completion, the input training file is deleted as it is no longer required. To monitor the training progress the *Get Model* status should be called repeatedly (see Upload File for Recommendation Type Training [page 46]). See Model Training [page 11] for a recommended approach on how to initiate model training.

Key	Value
End-Point	Training
HTTP Method	POST
Path	model/train
Headers	Content-Type: application/json Authorization: Bearer <token> X-Request-ID: <unique string>

Related Information

Input Attributes [page 84]
Expected Response [page 84]
Examples [page 84]

6.3.1 Input Attributes

The body should contain a JSON dictionary with the following entries:

Attribute	Type	Description
`model_id`	String	Unique identifier for the model.

6.3.2 Expected Response

The call returns a JSON dictionary with the following values:

Attribute	Type	Description
`model_id`	String	Unique identifier for the model.
`language`	String	The language code as specified during training file upload.
`status`	Integer	Function call status code. See Common Status and Error Codes [page 171] for possible values.
`status_message`	String	User-friendly error message.

6.3.3 Examples

Sample Request

Example

```
POST /sti/training/model/train HTTP/1.1
Authorization: Bearer ed195a25c62942aa85dce785d8d33496
X-Request-ID: D9D2D62BB5EA4EBD8E54F1EF16F5925C-2147483648-2147516416
{
    "model_id": "5836718c1925eb000dc6810e"
}
```

Example

Sample Responses

Positive Response

```
{
  "status": 0,
  "model_id": "5836718c1925eb000dc6810e",
  "status_message": "ok",
  "language": "en"
}
```

Negative Response: incremental model version is incompatible with runtime version

```
{
  "status": 74,
  "model_id": "5836718c1925eb0221dc6810e2",
  "status_message": "reference model version is incompatible"
}
```

Negative Response: incremental training reference model status is INVALID, IN_TRAINING, or NEW

```
{
  "status": 85,
  "model_id": "5836718c1925eb000dc1210e2",
  "status_message": "reference model status is not valid for incremental
training"
}
```

Negative Response: model not found

```
{
  "status": 92,
  "model_id": "5836718c1925eb000dc6810e2",
  "status_message": "model not found"
}
```

Negative Response: model not eligible for training (to start training, the model status must be NEW)

```
{
  "status": 95,
  "model_id": "5836718c1925eb000dc7710e2",
  "status_message": "unable to complete model training"
}
```

6.4 Activate Model

Activates a trained model which deactivates the current active model for the same type, language, and business object. Only models with status READY can be activated.

Model caching mechanism is being implemented in the inference services to reduce constant re-loading of same model and optimize throughput. Hence, the model is not loaded for inference service immediately after

the model activation, it queues for the next reload cycle (global setting approximated 180 seconds), during this intermediate state, the model status is LOADING.

Key	Value
End-Point	Training
HTTP Method	PUT
Path	model/activate
Headers	Content-Type: application/json Authorization: Bearer <token> X-Request-ID: <unique string>

Related Information

Input Attributes [page 86]
Expected Response [page 86]
Examples [page 87]

6.4.1 Input Attributes

The body should contain a JSON dictionary with the following entries:

Attribute	Type	Description
`model_id`	String	Unique identifier for the model.

6.4.2 Expected Response

The call returns a JSON dictionary with the following values:

Attribute	Type	Description
`model_id`	String	Unique identifier for the model.

Attribute	Type	Description
`language`	String	The language code as specified during training file upload.
`status`	Integer	Function call status code. See Common Status and Error Codes [page 171] for possible values.
`status_message`	String	User-friendly error message.

6.4.3 Examples

Sample Request

Example

```
PUT /sti/training/model/activate HTTP/1.1
Content-Type: application/json
Authorization: Bearer ed195a25c62942aa85dce785d8d33496
X-Request-ID: D9D2D62BB5EA4EBD8E54F1EF16F5925C-2147483648-2147516416
{
    "model_id": "5836718c1925eb000dc6810e"
}
```

Example

Sample Responses

Positive Response

```
{
  "language": "en",
  "model_id": "5836718c1925eb000dc6810e",
  "status": 0,
  "status_message": "ok"
}
```

Negative Response: incremental training reference model status is INVALID, IN_TRAINING, or NEW

```
{
  "status": 85,
  "model_id": "c5836718c1925eb000dc6810e",
  "status_message": "reference model status is not valid for incremental
training"
}
```

Negative Response: model not found

```
{
  "status": 92,
  "model_id": "5836718c1925eb000dc6810e2",
  "status_message": "model not found"
}
```

Negative Response: model is already ACTIVE or IN_TRAINING

```
{
  "status": 96,
  "model_id": "c5836718c1925eb000dc6810e",
  "status_message": "failed to activate model"
}
```

6.5 List All Models

Returns the list of all models available in the system.

Key	Value
End-Point	Training
HTTP Method	GET
Path	model
Headers	Authorization: Bearer <token> X-Request-ID: <unique string>

Related Information

Input Attributes [page 89]
Expected Response [page 90]
Examples [page 91]

6.5.1 Input Attributes

The body should contain a JSON dictionary with the following entries:

Attribute	Type	Description
`model_status`	String Enumeration	Status of the model. See Model Status [page 7] for model status values (case insensitive). Default list by all status. For example, *{{`sti_service_url`}}/sti/training/model*: lists all models and *{{`sti_service_url`}}/sti/training/model?model_status=ACTIVE*: lists only models with ACTIVE *model_status*.
`model_business_object`	String Enumeration	Scenario object's business object filtering for the model. See Scenario Object [page 23] for business object values (case insensitive). Default lists everything (if not specified). For example, *{{`sti_service_url`}}/sti/training/model?model_business_object=ticket* : lists only models with scenario *business_object* Ticket.
`model_type`	String Enumeration	Scenario object's type filtering for the model. See Scenario Object [page 23] for business object values (case insensitive). Default lists everything (if not specified). For example, *{{`sti_service_url`}}/sti/training/model?model_type=classification*: lists only models with scenario *type* Classification.

Attribute	Type	Description
`model_language`	String Enumeration	Scenario object's language filtering for the model. See Scenario Object [page 23] for model language values (case insensitive). Default lists everything (if not specified). For example, *{{`sti_service_url`}}/sti/training/model?model_language=en*: lists only models with scenario *language* en (English).

> **i Note**
>
> Input Attributes are optional for this call. They can be used to filter the list of models by one or multiple of the following attributes:
>
> - `model_status`
> - `model_business_object`
> - `model_type`
> - `model_language`
>
> If no input attribute is selected for this call, the default behaviour is to list all models without any filtering.

6.5.2 Expected Response

The call returns a JSON dictionary with the following values:

Attribute	Type	Description
`results`	Array of Model List Objects	A collection of model identifiers grouped by their respective model type, languages, and business objects. There is no order in this array. See Model List Object [page 91] for a description of each element in this array.
`status`	Integer	Function call status code. See Common Status and Error Codes [page 171] for possible values.
`status_message`	String	User-friendly error message.

Related Information

Model List Object [page 91]

6.5.2.1 Model List Object

The Model List object name in JSON is `results` (Array).

Attribute	Type	Description
`type`	String	The type of Machine Learning service as specified during training file upload. See Scenario Object [page 23].
`language`	String	The language code as specified during training file upload. See Scenario Object [page 23].
`business_object`	String	The type of business object to classify specified as part of the training scenario. See Scenario Object [page 23].
`model_id`	Array<String>	The collection of model ID for this list in no particular order.

6.5.3 Examples

Related Information

All Models [page 92]
List Active Model [page 93]
List Models (Filter by Model Language) [page 94]
List Models (Filter by Model Business Object) [page 95]
List Models (Filter by Invalid Model Business Object) [page 95]

6.5.3.1 All Models

Sample Request

Example

```
GET /sti/training/model HTTP/1.1
Authorization: Bearer ed195a25c62942aa85dce785d8d33496
X-Request-ID: D9D2D62BB5EA4EBD8E54F1EF16F5925C-2147483648-2147516416
```

Example

Sample Responses

Positive Response: the model ID is grouped by model type, business object and language


```
{
  "results": [
    {
      "model_id": [
        "5836718c1925eb000dc6810e",
        "5836718c1925eb021211867e",
        "5836718c1925eb001113311e"
      ],
      "language": "en",
      "business_object": "ticket",
      "type": "classification"
    },
    {
      "model_id": [
        "5836718c1925eb022dc6810e",
      ],
      "language": "de",
      "business_object": "ticket",
      "type": "classification"
    },
    {
      "model_id": [
        "5836718c1925eb023ec6810e",
      ],
      "language": "en",
      "business_object": "ticket",
      "type": "recommendation"
    },
    {
      "model_id": [
        "58296718c1925eb023ec6810e",
      ],
      "language": "en",
      "business_object": "ticket",
      "type": "recommendation"
    }
  ],
  "status": 0,
  "status_message": "ok"
```

```
}
```

Positive Response (no model found)

```
{
  "results": [],
  "status": 0,
  "status_message": "ok"
}
```

6.5.3.2 List Active Model

Sample Request

Example

```
GET /sti/training/model?model_status=ACTIVE HTTP/1.1
Authorization: Bearer ed195a25c62942aa85dce785d8d33496
X-Request-ID: D9D2D62BB5EA4EBD8E54F1EF16F5925C-2147483648-2147516416
```

Example

Sample Response

Positive Response: the model ID is grouped by model type, business object and language

```
{
  "results": [
    {
      "model_id": [
        "5836718c1925eb000dc6810e"
      ],
      "language": "en",
      "business_object": "ticket",
      "type": "classification"
    },
    {
      "model_id": [
        "5836718c1925eb022dc6810e"
      ],
      "language": "de",
      "business_object": "ticket",
      "type": "classification"
    },
    {
      "model_id": [
        "5836718c1925eb023ec6810e",
      ],
      "language": "en",
      "business_object": "ticket",
      "type": "recommendation"
```

```
    }
  ],
  "status": 0,
  "status_message": "ok"
}
```

6.5.3.3 List Models (Filter by Model Language)

Sample Request

Example

```
GET /sti/training/model?model_language=en HTTP/1.1
Authorization: Bearer ed195a25c62942aa85ede785d8d33496
X-Request-ID: D9D2D62BB5EA4EBD8E54F1EF1633925C-2147483648-2147516416
```

Example

Sample Response

Positive Response (only **model ID with specified language returns)**

```
{
  "results": [
    {
      "model_id": [
        "5836718c1925eb000dc6810e"
      ],
      "language": "en",
      "business_object": "ticket",
      "type": "classification"
    },
    {
      "model_id": [
        "5836718c1925eb023ec6810e",
      ],
      "language": "en",
      "business_object": "ticket",
      "type": "recommendation"
    }
  ],
  "status": 0,
  "status_message": "ok"
}
```

6.5.3.4 List Models (Filter by Model Business Object)

Sample Request

Example

```
GET /sti/training/model?model_business_object=ticket HTTP/1.1
Authorization: Bearer ed195a25c62942aa85ede785d8d33496
X-Request-ID: D9D2D62BB5EA4EBD8E54F1EF1633925C-2147483648-2147516416
```

Example

Sample Response

Positive Response (only model ID with specified language returns)

```
{
  "results": [
    {
      "model_id": [
        "5836718c1925eb000dc6810e"
      ],
      "language": "en",
      "business_object": "ticket",
      "type": "classification"
    },
    {
      "model_id": [
        "5836718c1925eb023ec6810e",
      ],
      "language": "en",
      "business_object": "ticket",
      "type": "recommendation"
    }
  ],
  "status": 0,
  "status_message": "ok"
}
```

6.5.3.5 List Models (Filter by Invalid Model Business Object)

Sample Request

Example

```
GET /sti/training/model? model_business_object=XYZ HTTP/1.1
Authorization: Bearer ed195a25c62942aa85ede785d8d33496
X-Request-ID: D9D2D62BB5EA4EBD8E54F1EF1633925C-2147483648-2147516416
```

Example

Sample Response

Positive Response (but no model ID returned because specified business object does not match)

```
{
  "results": [],
  "status": 0,
  "status_message": "ok"
}
```

6.6 Get Model Accuracy

Returns the validation results given a model identifier. Currently, the *Get Model Accuracy* service is valid only for scenario type Classification.

Key	Value
End-Point	Training
HTTP Method	GET
Path	model/accuracy
Headers	Authorization: Bearer <token> X-Request-ID: <unique string>

Related Information

Input Parameters [page 97]
Expected Response [page 97]
Examples [page 101]

6.6.1 Input Parameters

The parameters are given as URL parameters in the request with the following entries:

Parameter	Type	Description
`model_id`	String	Unique identifier of the model being queried.
`threshold`	Float (optional)	Confidence threshold input to retrieve results from model performance simulation.

6.6.2 Expected Response

The call returns a Model Accuracy object with the values detailed in Model Accuracy Object [page 97] and Field Validation Result Object [page 98]. Note that the field-level accuracy can be calculated from the respective confusion matrices. See Model Status Object [page 37] to obtain a general model accuracy number.

Related Information

Model Accuracy Object [page 97]

6.6.2.1 Model Accuracy Object

Attribute	Type	Description
`model_id`	String	Unique identifier for the model.
`model_status`	String Enumeration	The status of the model. See Model Status [page 7] for status values.

Attribute	Type	Description
`validation_results`	Array of Field Validation Result object	Contains the confusion matrices of each field obtained in the validation phase of model training. Contains average and individual precision, recall, and f1-score of the model. See Field Validation Result Object [page 98] for definition.
`validation_results_at_thresholds`	Array of Field Validation Result object	Contains average and individual precision, recall, and f1-score of the model given a confidence threshold. See Field Validation Result at Thresholds Object [page 99] for definition.
`status`	Integer	Function call status code. See Common Status and Error Codes [page 171] for possible values.
`status_message`	String	Human-readable error message.

Related Information

Field Validation Result Object [page 98]
Field Validation Result at Thresholds Object [page 99]

6.6.2.1.1 Field Validation Result Object

The Field Validation Result object name in JSON is `validation_results` (Array).

Attribute	Type	Description
`field`	String	The name of the field. This corresponds to an output field name defined during training. See Mapping Object [page 24].

Attribute	Type	Description
`confusion_matrix`	Confusion Matrix object	Contains the confusion matrix of the corresponding output field from the validation phase. The field-level validation accuracy can be derived from this object. See Confusion Matrix Object [page 6].
`precision`	Array of double	Contains the precision values of each category in the corresponding output field.
`recall`	Array of double	Contains the recall values of each category in the corresponding output field.
`f1_score`	Array of double	Contains the f1-score values of each category in the corresponding output field.
`average_precision`	Double	Contains the weighted average precision value of the corresponding output field.
`average_recall`	Double	Contains the weighted average recall value of the corresponding output field.
`average_f1_score`	Double	Contains the weighted average f1-score value of the corresponding output field.

6.6.2.1.2 Field Validation Result at Thresholds Object

The Field Validation Result at Thresholds object name in JSON is `validation_results_at_thresholds` (Array).

Attribute	Type	Description
`threshold`	Float	Confidence threshold value as specified in Input Parameters [page 97].
`validation_results`	Array of Field Validation Results object	Contains average precision, recall, and f1-score of the model. See Field Validation Results Object [page 100].

Related Information

Field Validation Results Object [page 100]

6.6.2.1.2.1 Field Validation Results Object

The field validation results object name in JSON is `validation_results` (array).

Attribute	Type	Description
`accuracy`	Double	The expected accuracy of the model under the adopted threshold.
`average_precision`	Double	Contains the weighted average precision value of the corresponding output field under the adopted threshold.
`average_recall`	Double	Contains the weighted average recall value of the corresponding output field under the adopted threshold.
`average_f1_score`	Double	Contains the weighted average f1-score value of the corresponding output field under the adopted threshold.
`confusion_matrix`	Confusion Matrix object	Contains the confusion matrix of the corresponding output field from the validation phase under the adopted threshold. The field-level validation accuracy can be derived from this object. See Confusion Matrix Object [page 6].
`f1_score`	Array of double	Contains the f1-score values of each category in the corresponding output field under the adopted threshold.
`field`	String	The name of the field. This corresponds to an output field name defined during training. See Mapping Object [page 24].
`precision`	Array of double	Contains the precision values of each category in the corresponding output field under the adopted threshold.
`probability_of_exceeding_threshold`	Double	Contains the probability of results having confidence value higher than the threshold.

Attribute	Type	Description
`recall`	Array of double	Contains the recall values of each category in the corresponding output field under the adopted threshold.

6.6.3 Examples

Sample Request

Example

```
GET /sti/training/model/accuracy?model_id=5836718c1925eb000dc6810e HTTP/1.1
Authorization: Bearer ed195a25c62942aa85dce785d8d33496
X-Request-ID: D9D2D62BB5EA4EBD8E54F1EF16F5925C-2147483648-2147516416
```

Example

Sample Responses

Response when model status is READY or ACTIVE:


```
{
   "model_id":"5836718c1925eb000dc6810e",
   "model_status":"READY",
   "validation_results":[
      {
         "field":"ticket_type",
         "confusion_matrix":{
            "labels":[
               "complaint",
               "request",
               "compliment"
            ],
            "values":[
               [
                  942,
                  156,
                  118
               ],
               [
                  8,
                  1006,
                  9
               ],
               [
                  46,
                  18,
                  996
```

```
            ]
         ],
         "precision":[
            0.42045454545454547,
            0.5405405405405406,
            0.6428571428571429
         ],
         "recall":[
            0.7115384615384616,
            0.35714285714285715,
            0.4576271186440678
         ],
         “f1_score”:[
            0.5285714285714285,
            0.4301075268817204,
            0.5346534653465347
         ],
         "average_precision":[
            0.5392962758232219
         ],
         "average_recall":[
            0.5029940119760479
         ],
         “average_f1_score”:[
            0.49770233680560577
         ]
      }
   },
   {
      "field":"routing_inbox",
      "confusion_matrix":{
         "labels":[
            "reservations",
            "product_support",
            "housekeeping",
            "public_relations"
         ],
         "values":[
            [
               512,
               421,
               632,
               79
            ],
            [
               9,
               1026,
               8,
               21
            ],
            [
               27,
               17,
               987,
               34
            ],
            [
               31,
               14,
               874,
               47
            ]
         ],
         "precision":[
            0.42045454545454547,
            0.5405405405405406,
            0.6428571428571429,
            0.4859392839184595
```

```
            ],
            "recall":[
               0.7115384615384616,
               0.35714285714285715,
               0.4576271186440678,
               0.5438295849293848
            ],
            “f1_score”:[
               0.5285714285714285,
               0.4301075268817204,
               0.5346534653465347,
               0.4859392986038851
            ],
            "average_precision":[
               0.5392962758232219
            ],
            "average_recall":[
               0.5029940119760479
            ],
            “average_f1_score”:[
               0.49770233680560577
            ]
         }
      }
   ],
   "status":0,
   "status_message":"ok"
}
```


Response when threshold of 0.7 is adopted:


```
{
   "model_id":"c54a58d31a614f3ab014303eccdcd2e1",
   "model_status":"READY",
   "status":0,
   "status_message":"ok",
   "validation_results":[
      {
         "average_f1_score":0.6481298429521334,
         "average_precision":0.7143844475380864,
         "average_recall":0.6593406593406593,
         "confusion_matrix":{
            "labels":[
               "complaint",
               "compliment",
               "request"
            ],
            "values":[
               [
                  29,
                  2,
                  2
               ],
               [
                  16,
                  11,
                  2
               ],
               [
                  8,
                  1,
                  20
               ]
            ]
         },
         "f1_score":[
            0.6744186046511628,
```

```
            0.5116279069767441,
            0.7547169811320755
         ],
         "field":"label",
         "precision":[
            0.5471698113207547,
            0.7857142857142857,
            0.8333333333333334
         ],
         "recall":[
            0.8787878787878788,
            0.3793103448275862,
            0.6896551724137931
         ]
      }
   ],
   "validation_results_at_thresholds":{
      "threshold":0.7,
      "validation_results":[
         {
            "accuracy":0.77,
            "average_f1_score":0.75,
            "average_precision":0.77,
            "average_recall":0.77,
            "confusion_matrix":{
               "labels":[
                  "complaint",
                  "compliment",
                  "request"
               ],
               "values":[
                  [
                     15,
                     2,
                     0
                  ],
                  [
                     6,
                     3,
                     0
                  ],
                  [
                     1,
                     0,
                     12
                  ]
               ]
            },
            "f1_score":[
               0.77,
               0.43,
               0.96
            ],
            "field":"label",
            "precision":[
               0.68,
               0.6,
               1.0
            ],
            "probability_of_exceeding_threshold":0.43,
            "recall":[
               0.88,
               0.33,
               0.92
            ]
         }
      ]
   }
```

```
}
```

Response when confusion matrix is not supported (currently only the scenario type Classification is supported):

```
{
  "model_id": "5836718c19aaA000dc6810e",
  model_status": "READY",
  "status": 75,
  "status_message": "confusion matrix not supported"
}
```

Response when the model is not yet trained (basically an empty response):

```
{
  "model_id": "5836718c1925eb000dc6810e",
  "model_status": "NEW",
  "validation_results": [
    {
      "field": "ticket_type",
      "confusion_matrix": []
    },
    {
      "field": "routing_inbox",
      "confusion_matrix": []
    }
  ],
  "status": 0,
  "status_message": "ok"
}
```

Response when the model is invalid:

```
{
  "model_id": "5836718c1925eb000dc6810e",
  "model_status": "INVALID",
  "status": 0,
  "status_message": "ok"
}
```

6.7 Delete Model or Training File

Removes a machine learning model or an uploaded training file to save storage space. Models that are ACTIVE (see Model Status [page 7]) and IN_TRAINING (for less than 50 hours) cannot be deleted. However, if the model status has remained IN_TRAINING for more than 50 hours and training is not yet completed successfully, it is possible that there has been an internal error, and the model can be deleted.

Key	Value
End-Point	Training
HTTP Method	DELETE

Key	Value
Path	model
Headers	Content-Type: application/json Authorization: Bearer <token> X-Request-ID: <unique string>

Related Information

Input Parameters [page 106]
Expected Response [page 106]
Examples [page 107]

6.7.1 Input Parameters

The parameters are given as URL parameters in the request with the following entries:

Parameter	Type	Description
`model_id`	String	Unique identifier for the model.

6.7.2 Expected Response

The call returns a JSON dictionary with the following values:

Attribute	Type	Description
`model_id`	String	Unique identifier for the model.
`status`	Integer	Function call status code. See Common Status and Error Codes [page 171] for possible values.
`status_message`	String	User-friendly error message.

6.7.3 Examples

Sample Request

Example

```
DELETE /sti/training/model?model_id=5836718c1925eb000dc6810e HTTP/1.1
Authorization: Bearer ed195a25c62942aa85dce785d8d33496
X-Request-ID: D9D2D62BB5EA4EBD8E54F1EF16F5925C-2147483648-2147516416
```

Example

Sample Responses

Positive response if the model status is NEW, READY, INVALID, or IN_TRAINING (more than 50 hours)

```
{
  "model_id": 5836718c1925eb000dc6810e",
  "status": 0,
  "status_message": "ok"
}
```

Negative response if the model status is ACTIVE, LOADING, or IN_TRAINING (less than 50 hours)

```
{
  "status": 97,
  "model_id": "58258537248d5d000f1492b7",
  "status_message": "failed to delete model"
}
```

6.8 Get Training Topics Tabulated

Returns clusters of tokens in the training data and how each token contributes to the selection of a prediction label. This can be used to determine the general topics of the data. Topic extraction is performed at model training and it is performed only on the input fields of the training data. Currently the *Get Training Topics Tabulated* service is valid only for the scenario type Classification.

Key	Value
End-Point	Training

Key	Value
HTTP Method	GET
Path	topic/tabulate
Headers	Content-Type: application/json Authorization: Bearer <token> X-Request-ID: <unique string>

Related Information

Input Parameters [page 108]
Expected Response [page 109]
Examples [page 110]

6.8.1 Input Parameters

The parameters are given as URL parameters in the request with the following entries:

Parameter	Type	Description
`model_id`	String	Unique identifier for the model.
`sources`	List of String (separated by comma)	The set of output sources to be returned. It should match one of the source names specified in Tabulating Object [page 28]. It returns all tabulation results by the source name when the source name is used in multiple category topic pairs.

6.8.2 Expected Response

The call returns a JSON dictionary with the following values:

Attribute	Type	Description
`model_id`	String	Unique identifier for the model.
`model_status`	String Enumeration	Status of the model. See Model Status [page 7] for model status values.
`results`	Array of Tabulation object.	Contains the topic tabulation object by output source. See Results Object [page 109] for definition.
`status`	Integer	Function call status code. See Common Status and Error Codes [page 171] for possible values.
`status_message`	String	User-friendly error message.

Related Information

Results Object [page 109]

6.8.2.1 Results Object

The Results object name in JSON is `results` (Array).

Attribute	Type	Description
`source`	String	The output source name.
`topic`	String	The output topic target name.
`tabulates`	Array of tabulates result object	Contains the tabulates for each output source and topic. See Tabulates Object [page 110] for definition.

Related Information

Tabulates Object [page 110]

6.8.2.1.1 Tabulates Object

The Tabulates object name in JSON is `tabulates` (Array).

Attribute	Type	Description
`label`	String	The output topic label.
`token_values`	Array of token value objects	It contains a list of tokens for the topic label and their respective weights. See Token Value Object [page 141] for definition. The length of the array is set to the maximum of 100 tokens per label.

6.8.3 Examples

Example

Sample Request

Request without sources field:

```
GET /sti/training/topics/tabulate?model_id=5836718c1925eb000dc6810e HTTP/1.1
Content-Type: application/json
Authorization: Bearer ed195a25c62942aa85dce785d8d33496
X-Request-ID: D9D2D62BB5EA4EBD8E54F1EF16F5925C-2147483648-2147516416
```

Request with sources field:

```
GET /sti/training/topics/tabulate?
model_id=5836718c1925eb000dc6810e&sources=body  HTTP/1.1
Content-Type: application/json
Authorization: Bearer ed195a25c62942aa85dce785d8d33496
X-Request-ID: D9D2D62BB5EA4EBD8E54F1EF16F5925C-2147483648-2147516416
```

Example

Sample Responses

Response when model status is READY, LOADING or ACTIVE:


```
{
  "model_id": "5836718c1925eb000dc6810e",
  "model_status": "READY",
  "results": [
    {
      "source": "body",
      "topic": "ticket_type",
      "tabulates": [
        {
          "label": "compliment",
          "token_values": [
            {
              "token": "Metropolis",
              "value": 0.91
            },
            {
              "token": "Gotham",
              "value": 0.91
            },
            {
              "token": "love",
              "value": 0.81
            },
            {
              "token": "Prestissimo",
              "value": 0.731
            },
            {
              "token": "upgraded",
              "value": 0.702
            },
            {
              "token": "comfy",
              "value": 0.62
            }
          ]
        },
        {
          "label": "complaint",
          "token_values": [
            {
              "token": "disappointment",
              "value": 0.85
            },
            {
              "token": "replacement",
              "value": 0.81
            },
            {
              "token": "lose",
              "value": 0.79
            },
            {
              "token": "waiting",
              "value": 0.74
            },
            {
              "token": "slower",
              "value": 0.71
            },
```

```
            {
              "token": "fought",
              "value": 0.63
            }
          ]
        }
      ]
    },
    {
      "source": "body",
      "topic": "routing_inbox",
      "tabulates": [
        {
          "label": "reservations",
          "token_values": [
            {
              "token": "Metropolis",
              "value": 0.91
            },
            {
              "token": "Gotham",
              "value": 0.85
            },
            {
              "token": "waiting",
              "value": 0.74
            },
            {
              "token": "fought",
              "value": 0.63
            }
          ]
        },
        {
          "label": "product_support",
          "token_values": [
            {
              "token": "love",
              "value": 0.81
            },
            {
              "token": "Prestissimo",
              "value": 0.731
            },
            {
              "token": "upgraded",
              "value": 0.702
            }
          ]
        },
        {
          "label": "public_relations",
          "token_values": [
            {
              "token": "Tan Baggins",
              "value": 0.603
            },
            {
              "token": "thank",
              "value": 0.62
            },
            {
              "token": "support",
              "value": 0.52
            }
          ]
        }
      ]
```

```
    }
   ],
  "status": 0,
  "status_message": "ok"
}
```

Response when model status is NEW or IN_TRAINING:

```
{
  "model_id": "5836718c1925eb000dc6810e",
  "model_status": "NEW",
  "results": [
    {
      "source": "body",
      "topic": "ticket_type",
      "tabulates": []
     },
    {
      "source": "body",
      "topic": "routing_inbox",
      "tabulates": []
    }
   ],
  "status": 0,
  "status_message": "ok"
}
```

Response when the model is invalid:

```
{
  "model_id": "5836718c1925eb000dc6810e",
  "model_status": "INVALID",
  "status": 0,
  "status_message": "ok"
}
```

6.9 Upload Incremental Training File / Data Extraction

Uploads incremental training file to create new ML models from existing ML models (reference model). The required inputs for incremental training are the reference model ID and the incremental training data. The reference model ID is used to retrieve model metadata from the reference model, and the ML configuration settings (such as scenario, field mapping, and topic tabulating) are inherited from the reference model. Hence the incremental training file contents should conform to the ML configuration settings of the reference model.

Upload incremental training file returns a new model ID, and the corresponding machine learning model can be created by calling Start Model Training [page 83] with this new model ID.

Note that the model count for the tenant includes the initial model as well as subsequent incremental models. The same conditions for maximum number of models and minimum / recommended number of samples apply for incremental training, as stated in Upload Training File / Data Extraction [page 20].

The Upload Incremental Training File endpoint also provides Data Extraction functionality. The Data Extraction process for Incremental Training is as described in Upload Training File / Data Extraction [page 20] . Like in upload incremental training file, ML configuration settings (such as scenario, field mapping, and topic

tabulating) are inherited from the reference model. Hence the OData information for incremental training should conform to the ML configuration settings of the reference model.

Key	Value
End-Point	Training
HTTP Method	POST
Path	Increment
Headers	Content-Type: application/json Authorization: Bearer <token> X-Request-ID: <unique string>

Related Information

Input Attributes [page 114]
Expected Response [page 115]
Examples [page 115]

6.9.1 Input Attributes

The body should contain a JSON dictionary with the following entries:

Attribute	Type	Description
`reference_model_id`	String	The unique identifier for the reference model. Ensure the reference model ID is valid.
`training`	Training Object (optional)	It uploads the file used as training data. See Training Object [page 32] for definition.
`odata`	OData Object (optional)	OData information to be used for data extraction. See OData Object [page 32] for definition. This object should be present only for data extraction functionality.

6.9.2 Expected Response

The call returns a Model Status object containing the model identifier (a summary of the training data). See Model Status Object [page 37] for definition.

6.9.3 Examples

Related Information

Upload Incremental File for Classification Type Training [page 115]
Upload Incremental File for Recommendation Type Training [page 117]
Data Extractor Type Training [page 118]

6.9.3.1 Upload Incremental File for Classification Type Training

Example

Sample Request (file upload)

```
POST /sti/training/increment HTTP/1.1
Content-Type: application/json
Authorization: Bearer ed195a25c62942aa85dce78668d33496
X-Request-ID: 440622E1-E0F4-4A60-A1A7-172090AA739A.2147483648.2147516416
{
  "reference_model_id": "5836718c1925eb000dc6810e",
  "training": {
    "file":
"c3ViamVjdCxib2R5LHRpY2tldF90eXBlLHJvdXRpbmdfaW5ib3gKamFja2V0IGZvdW5kLE1hc3NpdmUg
dGhhbmtzIHRvICBmb3IgZmluZGluZyBteSAgamFja2V0IChha2EgdGhlIGJlc3QgdHJhdmVsIHBpbGxvd
y9ibGFua2V0L2x1bWJhciBzdXBwb3J0IGV2ZXIpISxjb21wbGltZW50LHB1YmxpY19yZWxhdGlvbnMKa3
Vkb3MhLHNob3V0b3V0IHRvIGZsaWdodCBhdHRlbmRhbnQgQW5nZWxhIEFBIDI1NSBKRkstTEFYIHllc3R
lcmRheS4gU2hlIGhhbmRsZWQgbXkgZGF1Z2h0ZXJzIG1vdGlvbiBzaWNrbmVzcyBsaWtlIGEgcHJvLixj
b21wbGltZW50LHB1YmxpY19yZWxhdGlvbnMKdGhhbmtzISxMb3ZlIGdldHRpbmcgdW5leHBlY3RlZCB1c
GdyYWRlIGZvciB0aGUgcmlkZSBob21lLiBUaGFua3MgIChhdCAgICxjb21wbGltZW50LHB1YmxpY19yZW
xhdGlvbnM="
  }
}
```

Example

Sample Response (file upload)

Positive Response

```
{
  "model_id": "5836718c1925eb1212012778a12e",
  "model_status": "NEW",
  "desc": "Simple Twitter Incremental Training Data",
  "type": "classification",
  "business_object": "ticket",
  "language": "en",
  "tabulations_count": 2,
  "records_count": 3,
  "reference_model_id": "5836718c1925eb000dc6810e",
  "incremental": true,
  "incremental_upload_timestamp": "2017-05-17 04:56:30 UTC",
  "extracted_input": [
    {
      "type": "str",
      "field": "subject"
    },
    {
      "type": "str",
      "field": "body"
    }
  ],
  "extracted_output": [
    {
      "field": "ticket_type",
      "distribution": [{
          "value": "compliment", "count": 3
        }]
    },
    {
      "field": "routing_inbox",
      "distribution": [{
          "value": "public_relations", "count": 3
        }]
    }
  ],
  "status": 0,
  "status_message": "ok"
}
```

Negative Response if the reference model version is incompatible with runtime version

```
{
  "status": 74,
  "reference_model_id": "5836718c1925eb000dc681212",
  "status_message": " reference model version is incompatible"
}
```

Negative Response: incremental training reference model status is invalid

```
{
  "status": 84,
  "reference_model_id": "5836718c1925eb000dc681122",
  "status_message": "reference model status is not valid for incremental upload"
}
```

Negative Response if the reference model ID is not found

```
{
  "status": 92,
  "reference_model_id": "5836718c1925eb000dc6810e2",
  "status_message": "model not found"
}
```

6.9.3.2 Upload Incremental File for Recommendation Type Training

Example

Sample Request

```
POST /sti/training/increment HTTP/1.1
Content-Type: application/json
Authorization: Bearer ed195a25c62942aa85dce78668d33496
X-Request-ID: 440622E1-E0F4-4A60-A1A7-172090AA739A.2147483648.2147516416
{
  "reference_model_id": "5836718c1925eb000dc6810e",
  "training": {
    "file":
"c3ViamVjdCxib2R5LHRpY2tldF90eXBlLHJvdXRpbmdfaW5ib3gKamFja2V0IGZvdW5kLE1hc3NpdmUg
dGhhbmtzIHRvICBmb3IgZmluZGluZyBteSAgamFja2V0IChha2EgdGhlIGJlc3QgdHJhdmVsIHBpbGxvd
y9ibGFua2V0L2x1bWJhciBzdXBwb3J0IGV2ZXIpISxjb21wbGltZW50LHB1YmxpY19yZWxhdGlvbnMKa3
Vkb3MhLHNob3V0b3V0IHRvIGZsaWdodCBhdHRlbmRhbnQgQW5nZWxhIEFBIDI1NSBKRkstTEFYIHllc3R
lcmRheS4gU2hlIGhhbmRsZWQgbXkgZGF1Z2h0ZXJzIG1vdGlvbiBzaWNrbmVzcyBsaWtlIGEgcHJvLixj
b21wbGltZW50LHB1YmxpY19yZWxhdGlvbnMKdGhhbmtzISxMb3ZlIGdldHRpbmcgdW5leHBlY3RlZCB1c
GdyYWRlIGZvciB0aGUgcmlkZSBob21lLiBUaGFua3MgIChhdCAgICxjb21wbGltZW50LHB1YmxpY19yZW
xhdGlvbnM="
  }
}
```

Example

Sample Response

Positive response

```
  {
    "language": "en",
    "status_message": "ok",
    "type": "recommendation",
    "incremental_upload_timestamp": "2017-09-04 07:03:06 UTC",
    "desc": "Train solution recommender random questions",
    "incremental": true,
    "reference_model_id": "59acf7bb12a83d0007e7691d",
    "records_count": 1000,
    "status": 0,
```

```
    "model_status": "NEW",
    "extracted_input": [
        {
            "field": "subject",
            "type": "str"
        }
    ],
    "extracted_output": [
        {
            "field": "datasource"
        },
        {
            "field": "article_id"
        }
    ],
    "model_id": "5836718c1925eb000dc6810e",
    "business_object": "ticket"
}
```

Negative Response if the reference model version is incompatible with runtime version

```
{
  "status": 74,
  "reference_model_id": "5836718c1925eb000dc681212",
  "status_message": " reference model version is incompatible"
}
```

Negative Response: incremental training reference model status is invalid

```
{
  "status": 84,
  "reference_model_id": "5836718c1925eb000dc681122",
  "status_message": "reference model status is not valid for incremental upload"
}
```

Negative Response if the reference model ID is not found

```
{
  "status": 92,
  "reference_model_id": "5836718c1925eb000dc6810e2",
  "status_message": "model not found"
}
```

6.9.3.3 Data Extractor Type Training

Example

Data Extractor Sample Request (with template)

Sample Code

```
POST /sti/training/increment HTTP/1.1
Content-Type: application/json
Authorization: Bearer ed195a25c62942aa85dce785d8d33496
```

```
X-Request-ID: 440622E1-E0F4-4A60-A1A7-172090AA739A.2147483648.2147516416
{
   "reference_model_id": "5836718c1925eb000dc6810e",
   "odata": {
      "credential": {
         "service_url": "https://my******.crm.ondemand.com/sap/c4c/odata/v1/
c4codata",
         "authorization": "Basic ************************************"
      },
      "application": "C4C",
      "template": "ServiceRequest",
      "date_from": "2017-01-01T04:00:00.00Z"
   }
}
```

Example

Data Extractor Sample Request (without template)

Sample Code

```
POST /sti/training/increment HTTP/1.1
Content-Type: application/json
Authorization: Bearer ed195a25c62942aa85dce785d8d33496
X-Request-ID: 440622E1-E0F4-4A60-A1A7-172090AA739A.2147483648.2147516416
{
    "reference_model_id": "5836718c1925eb000dc6810e",
    "odata":{
        "credential":{
                  "service_url":"https://my******.crm.ondemand.com/sap/c4c/
odata/v1/c4codata",
    "authorization":"Basic ************************************"
    },
    "application":"C4C",
    "query_options":{
            "entity":"ServiceRequestCollection",
            "select":"Name,ServiceIssueCategoryID",
            "filter":"ServiceIssueCategoryID ne null and ETag ge
datetimeoffset'2017-01-01T04:00:00.00Z'",
            "orderby":"",
            "expand":"",
            "batch_query_options":{
                "entity":"ServiceRequestDescription",
                "select":"Text",
                "filter":"TypeCode eq '10004'",
                "orderby":"",
                "expand":""
            }
        }
    }
}
```

Example

Data Extractor Sample Response

Positive Response

Sample Code

```
{
   "model_id": "5954b0448e18510007f9ecac",
    "status_message": "ok",
    "status": 0
}
```

Negative Response (odata info should not be empty)

Sample Code

```
{
  "status": 43,
  "status_message": "odata info should not be empty"
}
```

Negative Response (odata credential should not be empty)

Sample Code

```
{
  "status": 44,
  "status_message": "odata credential should not be empty"
}
```

Negative Response (odata service url should not be empty)

Sample Code

```
{
  "status": 45,
  "status_message": "odata service URL should not be empty"
}
```

Negative Response (odata authorization info should not be empty)

Sample Code

```
{
  "status": 46,
  "status_message": "odata authorization info should not be empty"
}
```

Negative Response (odata application info is should not be empty)

Sample Code

```
{
  "status": 47,
  "status_message": "odata application info is should not be empty"
}
```

Negative Response (invalid application)

Sample Code

```
{
  "status": 48,
  "status_message": "invalid application"
}
```

Negative Response (invalid odata template type)

Sample Code

```
{
  "status": 49,
  "status_message": "invalid odata template type"
}
```

Negative Response (invalid date format)

Sample Code

```
{
  "status": 50,
  "status_message": "invalid date format"
}
```

Negative Response (either *date_from* or *date_to* is allowed)

Sample Code

```
{
  "status": 53,
  "status_message": "either date_from or date_to is allowed"
}
```

Negative Response (odata query options should not be empty or provide template)

Sample Code

```
{
  "status": 54,
  "status_message": "odata query options should not be empty or provide
template"
}
```

Negative Response (odata entity name should not be empty)

Sample Code

```
{
  "status": 55,
  "status_message": "odata entity name should not be empty"
}
```

Negative Response (odata select option should not be empty)

Sample Code

```
{
  "status": 56,
  "status_message": "odata select option should not be empty"
}
```

Negative Response (input/output mapping is not same as select option)

Sample Code

```
{
  "status": 57,
  "status_message": "input/output mapping is not same as select option"
}
```

Negative Response (data extractor call failed without response)

Sample Code

```
{
  "status": 59,
  "status_message": "data extractor call failed without response"
}
```

6.10 Disable Model Category

Disables categories of a trained modelin the Classification scenario type. The disable category is allowed only on models with model status ACTIVE, LOADING or READY.

The category class disabled in model is not eligible for inference. The changes are not immediate, and are only effective after the model is reloaded.

Key	Value
End-Point	Training
HTTP Method	PUT
Path	model/disable/category
Headers	Content-Type: application/json Authorization: Bearer <token> X-Request-ID: <unique string>

Related Information

Input Attributes [page 123]
Expected Response [page 123]
Examples [page 124]

6.10.1 Input Attributes

The body should contain a JSON dictionary with the following entries:

Attribute	Type	Description
`model_id`	String	Unique identifier for the model.
`contents`	Array of Field-Categories Objects	A list of field-category object to be enabled. See Value Field-Categories Object [page 42] for definition.

The field names in *field* needs to match the keys provided in the original training data. These keys are matched against the field names specified in the *output* attribute in Mapping Object [page 24].

The category classes in *class* need to match categories in the trained model *enabled_category_class*. See Model Status Object [page 37] for definition.

6.10.2 Expected Response

The call returns a JSON dictionary with the following values:

Attribute	Type	Description
`model_id`	String	Unique identifier for the model.
`status`	Integer	Function call status code. See Common Status and Error Codes [page 171] for possible values.
`status_message`	String	User-friendly error message.

6.10.3 Examples

Sample Request

Example

```
PUT /sti/training/model/disable/category HTTP/1.1
Content-Type: application/json
Authorization: Bearer ed195a25c62942aa85dce785d8d33496
X-Request-ID: D9D2D62BB5EA4EBD8E54F1EF16F5925C-2147483648-2147516416
{
    "model_id": "595220225ff57f000b5907b8",
    "contents": [
    {
      "category_class": [
        "Tool Development"
      ],
      "field": "label"
    }
  ]
}
```

Example

Sample Responses

Positive Response

```
{
  "language": "en",
  "model_id": "595220225ff57f000b5907b8",
  "status": 0,
  "status_message": "ok"
}
```

Negative Response: model not found

```
{
  "status": 92,
  "model_id": "5836718c1925eb000dc6810e2",
  "status_message": "model not found"
}
```

Negative Response: not a classification model

```
{
  "status": 61,
  "status_message": "not a classification model",
  "model_id": "59351cc934f7ad000608d82e"
}
```

Negative Response: incompatible model version

```
{
  "status": 65,
  "status_message": "incompatible model version",
  "model_id": "58f891654fe7ff000db82e3b"
}
```

Negative Response: model status is not ACTIVE or READY

```
{
  "status": 90,
  "status_message": "invalid model status",
  "model_id": "5954b0448e18510007f9ecac"
}
```

Negative Response: if the category field is not the output field of trained model

```
{
  "status": 62,
  "status_message": "invalid output field",
  "model_id": "595220225ff57f000b5907b8"
}
```

Negative Response: if the category class is not an existing class of trained model

```
{
  "status": 63,
  "status_message": "invalid category classes",
  "model_id": "595220225ff57f000b5907b8"
}
```

Negative Response: the requested category class is already disabled

```
{
  "status": 64,
  "status_message": "nothing to modify in category classes, request already
satisfied",
  "model_id": "595220225ff57f000b5907b8"
}
```

Negative Response: the requested model has too few category classes to be disabled

```
{
  "status": 66,
  "status_message": "number of enabled categories should be at least 2",
  "model_id": "5954b0448e18510007f9ecac"
}
```

6.11 Enable Model Category

Enables categories of a trained model with Classification scenario type. The enable category is allowed only on models with model status ACTIVE, LOADING or READY.

Only the category class enabled in model is eligible for inference. The changes are not immediate, and are only effective after the model is reloaded.

Key	Value
End-Point	Training
HTTP Method	PUT
Path	model/enable/category
Headers	Content-Type: application/json Authorization: Bearer <token> X-Request-ID: <unique string>

Related Information

Input Attributes [page 126]
Expected Response [page 127]
Examples [page 127]

6.11.1 Input Attributes

The body should contain a JSON dictionary with the following entries:

Attribute	Type	Description
`model_id`	String	Unique identifier for the model.
`contents`	Array of Field-Categories Objects	A list of field-category object to be enabled. See Value Field-Categories Object [page 42] for definition.

The field names in *field* needs to match the keys provided in the original training data. These keys are matched against the field names specified in the *output* attribute in Mapping Object [page 24].

The category classes in *class* need to match categories in the trained model *disabled_category_class*. See Model Status Object [page 37] for definition.

6.11.2 Expected Response

The call returns a JSON dictionary with the following values:

Attribute	Type	Description
`model_id`	String	Unique identifier for the model.
`status`	Integer	Function call status code. See Common Status and Error Codes [page 171] for possible values.
`status_message`	String	User-friendly error message.

6.11.3 Examples

Sample Request

Example

```
PUT /sti/training/model/enable/category HTTP/1.1
Content-Type: application/json
Authorization: Bearer ed195a25c62942aa85dce785d8d33496
X-Request-ID: D9D2D62BB5EA4EBD8E54F1EF16F5925C-2147483648-2147516416
{
    "model_id": "595220225ff57f000b5907b8",
    "contents": [
    {
      "category_class": [
        "Tool Development"
      ],
      "field": "label"
    }
  ]
}
```

Example

Sample Responses

Positive Response:

```
{
  "language": "en",
  "model_id": "595220225ff57f000b5907b8",
```

```
  "status": 0,
  "status_message": "ok"
}
```

Negative Response: model not found

```
{
  "status": 92,
  "model_id": "5836718c1925eb000dc6810e2",
  "status_message": "model not found"
}
```

Negative Response: not a classification model

```
{
  "status": 61,
  "status_message": "not a classification model",
  "model_id": "59351cc934f7ad000608d82e"
}
```

Negative Response: incompatible model version

```
{
  "status": 65,
  "status_message": "incompatible model version",
  "model_id": "58f891654fe7ff000db82e3b"
}
```

Negative Response: model status is not ACTIVE or READY

```
{
  "status": 90,
  "status_message": "invalid model status",
  "model_id": "5954b0448e18510007f9ecac"
}
```

Negative Response: if the category field is not the output field of trained model

```
{
  "status": 62,
  "status_message": "invalid output field",
  "model_id": "595220225ff57f000b5907b8"
}
```

Negative Response: if the category class is not an existing class of trained model

```
{
  "status": 63,
  "status_message": "invalid category classes",
  "model_id": "595220225ff57f000b5907b8"
}
```

Negative Response: the requested category class is already enabled

```
{
  "status": 64,
  "status_message": "nothing to modify in category classes, request already
satisfied",
  "model_id": "595220225ff57f000b5907b8"
}
```

6.12 Deactivate Model

Deactivates an active model and changes the model status to READY. Only models with model status ACTIVE or LOADING can be deactivated.

Model caching mechanism is being implemented in the inference services to reduce constant re-loading of same model and optimize throughput. Hence, the model is not loaded for inference service immediately after the model is deactivated, it queues for the next reload cycle (global setting approximated 180 seconds), during this intermediate state, the model status is LOADING.

Key	Value
End-Point	Training
HTTP Method	PUT
Path	model/deactivate
Headers	Content-Type: application/json Authorization: Bearer <token> X-Request-ID: <unique string>

Related Information

Input Attributes [page 129]
Expected Response [page 130]
Examples [page 130]

6.12.1 Input Attributes

The body should contain a JSON dictionary with the following entries:

Attribute	Type	Description
`model_id`	String	Unique identifier for the model.

6.12.2 Expected Response

The call returns a JSON dictionary with the following values:

Attribute	Type	Description
`model_id`	String	Unique identifier for the model.
`status`	Integer	Function call status code. See Common Status and Error Codes [page 171] for possible values.
`status_message`	String	User-friendly error message.

6.12.3 Examples

Sample Request

Example

```
PUT /sti/training/model/deactivate HTTP/1.1
Content-Type: application/json
Authorization: Bearer ed195a25c62942aa85dce785d8d33496
X-Request-ID: D9D2D62BB5EA4EBD8E54F1EF16F5925C-2147483648-2147516416
    {
        "model_id": "5954b0448e18510007f9ecac"
    }
```

Example

Sample Responses

Positive Response:

```
{
    "status_message": "ok",
    "model_id": "5954b0448e18510007f9ecac",
    "status": 0
}
```

Negative Response: deactivate a non-active model

```
{
    "status_message": "failed to deactivate model",
```

```
    "model_id": "5954b0448e18510007f9ecac",
    "status": 73
}
```

Negative Response: model not found

```
{
    "status_message": "model not found",
    "model_id": "5954b0448e18510007f9ecac_no_model",
    "status": 92
}
```

7 Classification API

This is the ML scenario type Classification.

Related Information

Classify Messages [page 132]

7.1 Classify Messages

Classifies incoming customer service messages (also known as support tickets). The input language is detected automatically by the machine learning algorithm, but it can also be overridden by explicitly specifying the *language* input parameter (see Classification API Input Attributes [page 133]) in case the language detection capability is not satisfactory.

Key	Value
End-Point	Classification
HTTP Method	POST
Path	text/classify
Headers	Content-Type: application/json Authorization: Bearer <token> X-Request-ID: <unique string>

Related Information

Classification API Input Attributes [page 133]
Classification API Expected Response [page 138]
Classification API Examples [page 141]

7.1.1 Classification API Input Attributes

The body should contain a JSON dictionary with the following entries:

Attribute	Type	Description
`business_object`	String	The business object name. This should have been specified as part of the training data. See Scenario Object [page 23] for definition.
`language`	String (optional)	Denotes the language of the inference request contents. The language is granted during customer on-boarding (provisioning). This attribute is optional. If not specified, the language is detected automatically. However, if the detected language is not supported, it defaults to English.
`messages`	Array of Message Objects	Contains the sample tickets to be classified.
`model_id` **i Note** The `model_id` attribute has been deprecated. This feature is now available via parameter in the URL: **`{{sti_service_url}}/sti/text/classify?model_id=<model ID>`**. For example, if you want to use model **`12345`** for inference, make a call to: **`{{sti_service_url}}/sti/text/classify?model_id=12345`**. The response body will contain the usual inference request except the `model_id` since it is already present in the URL.	String (optional)	Specifies the model to use for prediction. When this is specified, the fields `language` and `business_object` are ignored. This field is used to test a trained model without activating it, hence the corresponding model status has to be either **READY** or **ACTIVE**.

Attribute	Type	Description
`options`	Classification Options Object (optional)	Describes the various classification options. See Classification Options Objects [page 135] for definition.

Related Information

Message Object [page 134]
Classification Options Objects [page 135]

7.1.1.1 Message Object

The Message object name in JSON is `messages` (Array).

Attribute	Type	Description
`id`	Integer	Unique identifier for the ticket/message.
`contents`	Array of Message Content objects	Each Message Content object contains the text for every field in the message. See Message Content Object [page 134] for definition.

Related Information

Message Content Object [page 134]

7.1.1.1.1 Message Content Object

A Message Content object encloses a single field comprising the message. For example, these fields could be the subject of an e-mail or its body. These fields are defined in the training phase of the model.

The Message Content object name in JSON is `contents` (Array).

Attribute	Type	Description
`field`	String	The name of the field. This needs to match one of the field names given in the input attribute in Mapping Object [page 24].
`value`	String	The free-form text contents of the field.

The field names in *fields* need to match the keys provided in the original training data. These keys are matched against the field names specified in the *input* attribute in Mapping Object [page 24]. Field names given here that do not match the training data fields are ignored. Similarly, fields not present in this array are assumed to be empty strings.

7.1.1.2 Classification Options Objects

The Classification Options Object specifies additional machine-learning functions to be performed on each message.

The Classification Options object name in JSON is `options`.

Attribute	Type	Description
`classification_top_k`	Int (optional)	When specified, returns the classification's top-k results as part of the results. Minimum value = 1, maximum value = 5 or the total number of category class (which ever come first).
`classification_keyword_top_k`	Int (optional)	When specified, returns the classification's top-k keywords as part of the results. Minimum value = 3, maximum value = 20 or the total number of words in classification message (which ever come first). This can be used to provide justification of the machine-learning output to the user.
`thresholds`	Array of Threshold field-value Object (optional)	The list of output field and threshold value, which indicates threshold will be applied to. Default behavior is that no threshold is applied if it is not provided. See Threshold Field-Value Object [page 136] for definition.

Attribute	Type	Description
`services`	Services object (optional)	Describes the various services options for the input request. Default values are used.

Related Information

Threshold Field-Value Object [page 136]
Services Object [page 136]

7.1.1.2.1 Threshold Field-Value Object

A Threshold field-value object encloses a single field and the respective threshold value to be applied.

The Threshold Field-Value object name in JSON is `thresholds` (Array).

Attribute	Type	Description
`field`	String	The name of the field, which indicates the field that threshold will be applied to.
`value`	Float	The value of threshold should be between 0.0 to 1.0.
`low_confidence_label`	String (optional)	The label that will be returned for classification results with confidence score lower than the threshold. If not provided, the default label *low confidence* is used.

7.1.1.2.2 Services Object

The same endpoint (Classify Message) is used to provide both category classification and language detection services. By default, if services option is not specified, only category classification is provided. Additional options can be set in the request payload, if required to include other services.

Attribute	Type	Description
`detect_category`	Boolean (optional)	When specified, the request performs category classification.
`detect_language`	Boolean (optional)	When specified, the request performs language detection.
`detect_sentiment`	Boolean (optional)	When specified, the request performs sentiment detection. The sentiment score for a given text is returned as `overall_score`. The `overall_score` can vary from -1 to 1, where -1 corresponds to highly negative sentiment and +1 corresponds to highly positive sentiment. 0 is usually considered as text with neutral emotion, but in cases where the input contains texts in both polarities they can offset the `overall_score` to 0. **! Restriction** This attribute is not available in the Google Cloud region US Central (IA).

Default Value

```
{
    "detect_sentiment": false,
    "detect_category": true,
    "detect_language": false
}
```

7.1.2 Classification API Expected Response

The call returns a JSON dictionary with the following values:

Attribute	Type	Description
`results`	Array of Message Classification Object	The resulting classification of messages object. Note that the order may differ from the one given in the request. See Message Classification Object [page 138] for definition.

Related Information

Message Classification Object [page 138]

7.1.2.1 Message Classification Object

The Message Classification object name in JSON is `results` (Array).

Attribute	Type	Description
`id`	Integer	The identifier of the original message. This is the same value given earlier as the `id` attribute in Message Object [page 134].
`classification`	Array of Classification Result objects	The primary classification results for each output field are expressed here. See Classification Result Object [page 139] for definition.
`keyword`	Array of Classification Keyword objects	Helps explain the classification output to the user. See Classification Keyword Object [page 140] for definition. Only available when the option `classification_keyword_top_k` is specified. See Classification Options Objects [page 135].

Attribute	Type	Description
`detected_language`	String	The detected language for all fields expressed in a two-letter language code (for example, *EN* for English and *DE* for German).
`status`	Integer	Status code of the given ticket's classification, which would be *0* when successful. If there is only a model for the English language but the message is detected as German, then the status would be *91*. See Common Status and Error Codes [page 171].
`status_message`	String	User-friendly error message.

Related Information

Classification Result Object [page 139]
Classification Keyword Object [page 140]
Token Value Object [page 141]

7.1.2.1.1 Classification Result Object

A Classification Result object provides information on a single classification label output.

The Classification Result object name in JSON is `classification` (Array).

Attribute	Type	Description
`field`	String	Contains the output field name of the classification. This would match one of the field names given by the corresponding mapping object's output attribute as specified when the model was trained See Expected Response [page 127].

Attribute	Type	Description
`value`	String	The predicted classification labels. If prediction confidence is lower than threshold option specified value, the returned label would be low confidence label provided by user, or default to `low confidence` if not provided.
`confidence`	Float	The confidence level of the label value above. It ranges from 0.0 to 1.0.
`top_k_value`	Array of Classification Result objects	See Classification Result Object for Top-k Value [page 140] for definition of Classification Result object. Available only when option `classification_top_k` is specified. See Classification Options Objects [page 135].

Related Information

Classification Result Object for Top-k Value [page 140]

7.1.2.1.1.1 Classification Result Object for Top-k Value

A Classification Result object encloses a single classification label and the respective confidence score.

The Classification Result object name in JSON is `top_k_value` (Array).

Attribute	Type	Description
`confidence`	Float	The confidence level of the label value below. It ranges from 0.0 to 1.0.
`value`	String	The classified labels.

7.1.2.1.2 Classification Keyword Object

A Classification Keyword object helps the user understand why the machine-learning algorithm chooses a particular classification label.

The Classification Keyword object name in JSON is `keyword` (Array).

Attribute	Type	Description
`field`	String	Contains the `input field` name of the classification. This would match one of the field names given by the corresponding mapping object's *input* attribute as specified when the model was trained. See Expected Response [page 127].
`token_values`	Array of Token Value objects	Contains a list of tokens and their respective weights. See Token Value Object [page 141] for definition.

7.1.2.1.3 Token Value Object

A Token Value object associates a rational value of the token extracted from the corresponding input field. Greater value means that the associated token is more important.

The Token Value objec name in JSON is `token_values` (Array).

Attribute	Type	Description
`token`	String	Contains the token extracted from the input text.
`value`	Float	Denotes the importance of the respective token. It ranges from 0.0 to 1.0.

7.1.3 Classification API Examples

Related Information

Classification with No Options [page 142]
Classification with Top-k Values [page 148]
Classification with Language Detection [page 150]
Classification with Sentiment Detection [page 151]
Classification with Sentiment, Language and Category Detection [page 152]

Classification with Multiple Fields in the Same Request [page 153]
Classification with Multiple Services in the Same Request [page 154]

7.1.3.1 Classification with No Options

Example

Sample classification request with no options:

```
POST /sti/text/classify HTTP/1.1
Content-Type: application/json
Authorization: Bearer ed195a25c62942aa85dce785d8d33496
X-Request-ID: D9D2D62BB5EA4EBD8E54F1EF16F5925C-2147483648-2147516416
{
  "business_object": "ticket",
  "messages": [
    {
      "id": 1001,
      "contents": [
        {
          "field": "subject",
          "value": "Been told I would get chance of fought confirmation by
@SkyletteAirlines after 24 hours and I'm still waiting. Not good enough"
        }
      ]
    },
    {
      "id": 1002,
      "contents": [
        {
          "field": "subject",
          "value": "@askSkyletteAirlines @SkyletteAirlines Well thanks very much
for making the check-in experience in SPL so ONEROUS. It's slower than the old
T1."
        }
      ]
    },
    {
      "id": 1003,
      "contents": [
        {
          "field": "subject",
          "value": "@HomerJSimpson @Prestissimo I love Prestissimo too... We
have just upgraded - love it!"
        }
      ]
    },
    {
      "id": 1004,
      "contents": [
        {
          "field": "subject",
          "value": "vacation"
        },
        {
          "field": "body",
          "value": "done. thank you for following up. please let me know when
someone is coming. please tell them to avoid further disappointment by bringing
```

```
the replacement with them. i do not want to have to reply saying that you guys
removed the device for another 2 weeks!"
        }
      ]
    },
    {
      "id": 1005,
      "contents": [
        {
          "field": "subject",
          "value": "Super comfy flight from Metropolis - Gotham with Premium
Flatbed by @SkyletteAirlines @SkyletteAirlinesPU #kbbltv… https://example.com/
5552135"
        }
      ]
    },
    {
      "id": 1006,
      "contents": [
        {
          "field": "subject",
          "value": "We would like to thank Tan Baggins @FrodoBaggins , Group
Chief Executive Officer of @SkyletteAirlines for his kind support.... https://
example.com/5551234"
        }
      ]
    },
    {
      "id": 1007,
      "contents": [
        {
          "field": "subject",
          "value": "Clexit - die syldavische Fotograf Tobias Funke und seine
Kampagne gegen einen USE-Ausstieg der Caledonen; Ungleichheit - caledonische
Langzeitstudie mit 5000 Probanden; Straenbettler - Geld geben"
        }
      ]
    }
  ],
  "options": {
    "classification_keyword": true,
    "thresholds": [
         {
       "field": "ticket_type",
         "value": 0.5,
         "low_confidence_label": "customer low confidence label"
         },
       {
       "field": "routing_inbox",
       "value": 0.53
       }
     ]
  }
}
```


Example

Sample classification response with no options:


```
{
  "results": [
    {
      "detected_language": "de",
```

```
      "id": 1007,
      "status": 91,
      "status_message": "no active model found for this language"
    },
    {
      "classification": [
        {
          "field": "ticket_type",
          "value": "complaint",
          "confidence": 0.87
        },
        {
          "field": "routing_inbox",
          "value": "reservations",
          "confidence": 0.73
        }
      ],
      "keyword": [
        {
          "field": "subject",
          "token_values": [
            {
              "token": "waiting",
              "value": 0.74
            },
            {
              "token": "fought",
              "value": 0.63
            }
          ]
        },
        {
          "field": "body",
          "token_values": []
        }
      ],
      "detected_language": "en",
      "id": 1001,
      "status": 0,
      "status_message": "ok"
    },
    {
      "classification": [
        {
          "field": "ticket_type",
          "value": "complaint",
          "confidence": 0.87
        },
        {
          "field": "routing_inbox",
          "value": "reservations",
          "confidence": 0.73
        }
      ],
      "keyword": [
        {
          "field": "subject",
          "token_values": [
            {
              "token": "ONEROUS",
              "value": 0.82
            },
            {
              "token": "slower",
              "value": 0.71
            },
            {
              "token": "check-in",
```

```
              "value": 0.64
            },
            {
              "token": "T1",
              "value": 0.34
            }
          ]
        },
        {
          "field": "body",
          "token_values": []
        }
      ],
      "detected_language": "en",
      "id": 1002,
      "status": 0,
      "status_message": "ok"
    },
    {
      "classification": [
        {
          "field": "ticket_type",
          "value": "compliment",
          "confidence": 0.89
        },
        {
          "field": "routing_inbox",
          "value": "product_support",
          "confidence": 0.78
        }
      ],
      "keyword": [
        {
          "field": "subject",
          "token_values": [
            {
              "token": "love",
              "value": 0.81
            },
            {
              "token": "Prestissimo",
              "value": 0.731
            },
            {
              "token": "upgraded",
              "value": 0.702
            }
          ]
        },
        {
          "field": "body",
          "token_values": []
        }
      ],
      "detected_language": "en",
      "id": 1003,
      "status": 0,
      "status_message": "ok"
    },
    {
      "classification": [
        {
          "field": "ticket_type",
          "value": "complaint",
          "confidence": 0.97
        },
        {
          "field": "routing_inbox",
```

```
          "value": "housekeeping",
          "confidence": 0.934
        }
      ],
      "keyword": [
        {
          "field": "subject",
          "token_values": [{
              "token": "vacation",
              "value": 1
            }]
        },
        {
          "field": "body",
          "token_values": [
            {
              "token": "disappointment",
              "value": 0.85
            },
            {
              "token": "replacement",
              "value": 0.81
            },
            {
              "token": "device",
              "value": 0.79
            },
            {
              "token": "comfy",
              "value": 0.62
            }
          ]
        }
      ],
      "detected_language": "en",
      "id": 1004,
      "status": 0,
      "status_message": "ok"
    },
    {
      "classification": [
        {
          "field": "ticket_type",
          "value": "compliment",
          "confidence": 0.97
        },
        {
          "field": "routing_inbox",
          "value": "reservations",
          "confidence": 0.934
        }
      ],
      "keyword": [
        {
          "field": "subject",
          "token_values": [
            {
              "token": "comfy",
              "value": 0.62
            },
            {
              "token": "Metropolis",
              "value": 0.91
            },
            {
              "token": "Gotham",
              "value": 0.91
            }
```

```
          ]
        },
        {
          "field": "body",
          "token_values": []
        }
      ],
      "detected_language": "en",
      "id": 1005,
      "status": 0,
      "status_message": "ok"
    },
    {
      "classification": [
        {
          "field": "ticket_type",
          "value": "complient",
          "confidence": 0.84
        },
        {
          "field": "routing_inbox",
          "value": "public_relations",
          "confidence": 0.76
        }
      ],
      "keyword": [
        {
          "field": "subject",
          "token_values": [
            {
              "token": "Tan Baggins",
              "value": 0.603
            },
            {
              "token": "thank",
              "value": 0.62
            },
            {
              "token": "support",
              "value": 0.52
            }
          ]
        },
        {
          "field": "body",
          "token_values": []
        }
      ],
      "detected_language": "en",
      "id": 1006,
      "status": 0,
      "status_message": "ok"
    }
  ]
}
```


Example

Sample classification response with no options and low confidence:


```
{
  "results": [
    {
```

```
      "classification": [
        {
          "field": "ticket_type",
          "value": "customer low confidence label",
          "confidence": 0.22
        },
        {
          "field": "routing_inbox",
          "value": "low confidence",
          "confidence": 0.13
        }
      ],
      "detected_language": "en",
      "id": 1007,
      "status": 0,
      "status_message": "ok"
    }
  ]
}
```

7.1.3.2 Classification with Top-k Values

Example

Sample classification request with top-k values:

```
{
  "business_object": "ticket",
  "language": "en",
  "messages": [
  {
    "id": 1001,
    "contents": [
    {
      "field": "TEXT",
       "value": "Super comfy flight from Metropolis - Gotham with Premium
Flatbed, good service, thanks"
    }]
  }
  ],
  "options": {
   "classification_top_k": 5,
   "classification_keyword": true,
   "classification_keyword_top_k": 10,
   "thresholds": [
       {
           "field": "ERMS CAT1",
           "value": 0.80,
           "low_confidence_label": "not confidence"
       }]
  }
}
```

Example

Sample classification response with top-k values:


```
{
  "results": [
    {
      "id": 1001,
      "detected_language": "en",
      "status": 0,
      "classification": [
        {
          "top_k_value": [
            {
              "confidence": 0.9860512018203735,
              "value": "ERMS COMPLIMENT"
            },
            {
              "confidence": 0.011740081012248993,
              "value": "ERMS COMPLAINT"
            },
            {
              "confidence": 0.00220871577039361,
              "value": "ERMS REQUEST"
            }
          ],
          "confidence": 0.9860512018203735,
          "value": "ERMS COMPLIMENT",
          "field": "ERMS CAT1"
        }
      ],
      "keyword": [
        {
          "token_values": [
            {
              "token": "thanks",
              "value": 0.09147345274686813
            },
            {
              "token": "Flatbed",
              "value": 0.08825122565031052
            },
            {
              "token": "service",
              "value": 0.08815401047468185
            },
            {
              "token": "Super",
              "value": 0.08718737959861755
            },
            {
              "token": "Gotham",
              "value": 0.08583559840917587
            },
            {
              "token": "Premium",
              "value": 0.08474037051200867
            },
            {
              "token": "comfy",
              "value": 0.08261019736528397
            },
            {
              "token": "from",
              "value": 0.08202414214611053
            },
```

```
            {
              "token": "flight",
              "value": 0.07854945212602615
            },
            {
              "token": "with",
              "value": 0.07796038687229156
            }
          ],
          "field": "TEXT"
        }
      ],
      "status_message": "ok"
    }
  ]
}
```


7.1.3.3 Classification with Language Detection

Example

Sample classification request with language detection:


```
{
    "business_object": "ticket",
    "messages": [
        {
            "id": 2001,
            "contents": [
                {
                    "field": "description",
                    "value": "¿Cuál es el horario de atención al cliente?"
                }
            ]
        }
    ],
    "options": {
        "services": {
            "detect_language": true
        }
    }
}
```


Example

Sample classification response with language detection:


```
{
    "results": [
        {
            "detected_language": "es",
            "id": 2001,
```

```
            "status": 0,
            "status_message": "ok"
        }
    ]
}
```

7.1.3.4 Classification with Sentiment Detection

Example

Sample classification request with sentiment detection:

```
{
    "business_object": "ticket",
    "messages": [
        {
            "id": 2001,
            "contents": [
                {
                    "field": "description",
                    "value": "The ticket booking experience is very frustrating"
                }
            ]
        }
    ],
    "options": {
        "services": {
            "detect_sentiment": true
        }
    }
}
```

Example

Sample classification response with sentiment detection:

```
{
    "results": [
        {
            "id": 2001,
            "sentiment": {
                "overall_score": -0.8
            },
            "status": 0,
            "status_message": "ok"
        }
    ]
}
```

7.1.3.5 Classification with Sentiment, Language and Category Detection

Example

Sample classification request with sentiment, language and category detection:

```
{
    "business_object": "ticket",
    "messages": [
        {
            "id": 2001,
            "contents": [
                {
                    "field": "description",
                    "value": "The ticket booking experience is very frustrating"
                }
            ]
        }
    ],
    "options": {
        "services": {
            "detect_sentiment": true,
            "detect_language": true,
            "detect_category": true
        }
    }
}
```

Example

Sample classification response with sentiment, language and category detection:

```
{
  "results": [
    {
      "classification": [
        {
          "confidence": 0.42738021056920605,
          "field": "category",
          "value": "complaint"
        }
      ],
      "detected_language": "en",
      "id": 2001,
      "sentiment": {
        "overall_score": -0.8
      },
      "status": 0,
      "status_message": "ok"
    }
  ]
}
```

7.1.3.6 Classification with Multiple Fields in the Same Request

Example

Sample classification request with multiple fields in the same request:

```
{
    "business_object": "ticket",
    "messages": [
        {
            "id": 2001,
            "contents": [
                {
                    "field": "description",
                    "value": "¿Cuál es el horario de atención al cliente?"
                },
                {
                    "field": "subject",
                    "value": "consulta general"
                }
            ]
        }
    ],
    "options": {
        "services": {
            "detect_language": true,
            "detect_category": true
        }
    }
}
```

Example

Sample classification response with multiple fields in the same request:

```
{
    "results": [
        {
            "detected_language": "es",
            "id": 2001,
            "classification": [
                {
                    "field": "ticket_type",
                    "value": "enquiry",
                    "confidence": 0.84
                }
            ],
            "status": 0,
            "status_message": "ok"
        }
    ]
}
```

7.1.3.7 Classification with Multiple Services in the Same Request

Example

Sample classification request with multiple services in the same request:

```
{
    "business_object": "ticket",
    "messages": [
        {
            "id": 2001,
            "contents": [
                {
                    "field": "description",
                    "value": "¿Cuál es el horario de atención al cliente?"
                }
            ]
        }
    ],
    "options": {
        "services": {
            "detect_language": true,
            "detect_category": true
        }
    }
}
```

Example

Sample classification response with multiple services in the same request:

```
{
    "results": [
        {
            "detected_language": "es",
            "id": 2001,
            "classification": [
                {
                    "field": "ticket_type",
                    "value": "enquiry",
                    "confidence": 0.84
                }
            ],
            "status": 0,
            "status_message": "ok"
        }
    ]
}
```

8 Recommendation API

This is the ML scenario type Recommendation.

Related Information

Recommend Solutions [page 155]

8.1 Recommend Solutions

Recommends possible solutions for incoming customer service messages (also known as support tickets). The input language is detected automatically by the machine-learning algorithm, but it can also be overridden by explicitly specifying the language input parameter (see Recommendation API Input Attributes [page 156]) in case the language detection capability is not satisfactory.

Key	Value
End-Point	Recommendation
HTTP Method	POST
Path	solution/recommend
Headers	Content-Type: application/json Authorization: Bearer <token> X-Request-ID: <unique string>

Related Information

Recommendation API Input Attributes [page 156]
Recommendation API Expected Response [page 158]
Recommendation API Examples [page 160]

8.1.1 Recommendation API Input Attributes

The body should contain a JSON dictionary with the following entries:

Attribute	Type	Description
`business_object`	String	The business object name. This should have been specified as part of the training data. See Scenario Object [page 23] for definition.
`language`	String (optional)	Denotes the language of the inference request contents. The language is granted during customer on-boarding (provisioning). This attribute is optional. If not specified, the language is detected automatically. However, if the detected language is not supported, it defaults to English.
`messages`	Array of Message Objects	Contains the sample tickets to be recommended. See Message Object [page 134] for definition.
`model_id`	String (optional)	Specifies the model to use for prediction. When this is specified, the fields `language` and `business_object` are ignored. This field is used to test a trained model without activating it, hence the corresponding model status has to be either **READY** or **ACTIVE**. Furthermore, the tenant needs to have at least one other recommendation model that is already **ACTIVE** or otherwise the prediction may fail.
`options`	Solution Recommendation Options Object (optional)	Describes the various solution recommendation options. See Solution Recommendation Options Object [page 157] for definition.

Related Information

Solution Recommendation Options Object [page 157]
Threshold Value Object [page 157]

8.1.1.1 Solution Recommendation Options Object

The Solution Recommendation Options object specifies additional machine-learning functions to be performed on each message. Currently the only Solution Recommendation Option supported is `threshold`.

The Solution Recommendation Options object name in JSON is `options`.

Attribute	Type	Description
`threshold`	Threshold Value Object (optional)	Threshold value and associated high and low confidence label. Default behavior is that no threshold is applied if it is not provided. See Threshold Value Object [page 157] for definition.
`recommendation_top_k`	Integer (optional)	The value of `recommendation_top_k` must be an integer and it indicates how many results are required to be outputted from the model.

8.1.1.2 Threshold Value Object

A Threshold Value object encloses the threshold value and customizable high or low confidence label to be applied.

The Threshold Value object name in JSON is `threshold`.

Attribute	Type	Description
`value`	Float	The value of threshold should be between 0.0 to 1.0.
`low_confidence_label`	String (optional)	The label that will be returned for recommendation result with confidence score lower than the threshold. If not provided, default label *low* is used.
`high_confidence_label`	String (optional)	The label that will be returned for recommendations with confidence score higher than the threshold. If not provided, default label *high* is used.

8.1.2 Recommendation API Expected Response

The call returns a JSON dictionary with the following values:

Attribute	Type	Description
`results`	Array of Message Recommendation Object	The recommended solutions for messages object. Note that the order may differ from the one given in the request. See Message Recommendation Object [page 158] for definition.

Related Information

Message Recommendation Object [page 158]

8.1.2.1 Message Recommendation Object

The Message Recommendation object name in JSON is `results` (Array).

Attribute	Type	Description
`id`	Integer	The identifier of the original message. This will be the same value given earlier as the *id* attribute of a Message Object. See Message Object [page 134].
`recommendation`	Array of Recommendation Result objects	Contains a list of results for solution recommendation and their respective scores. See Recommendation Result Object [page 159] for definition.
`detected_language`	String	The detected language for all fields expressed in a two-letter language code (*EN* for English and *DE* for German, for example).

Attribute	Type	Description
`status`	Integer	Status code of the given ticket's recommendation, which would be *0* when successful. If there is only a model for the English language but the message is detected as German, then the status would be *91*. See Common Status and Error Codes [page 171].
`status_message`	String	User-friendly error message.

Related Information

Recommendation Result Object [page 159]

8.1.2.1.1 Recommendation Result Object

A Recommendation Result object provides information on a single solution recommendation output.

The Recommendation Result object name in JSON is `recommendation` (Array).

Attribute	Type	Description
`score`	Float	The similarity score of the solution recommendation that ranges between 0 to 1.
`solutions`	Array of Solution Results	Each output field of the solution recommendation is expressed here. See Solution Result Object [page 160] for definition.

Related Information

Solution Result Object [page 160]

8.1.2.1.1.1 Solution Result Object

A Solution Result object provides information on a single recommendation output.

The Solution Result object name in JSON is `solutions` (Array).

Attribute	Type	Description
`field`	String	Contains the output field name of the solution recommendation. This should match one of the field names given by the corresponding recommending object's `output` attribute as specified when the model was trained in Mapping Object [page 24].
`value`	String	The solution recommendation result.

8.1.3 Recommendation API Examples

Example

Sample Request (no options specified):

```
POST /sti/solution/recommend HTTP/1.1
Content-Type: application/json
Authorization: Bearer ed195a25c62942aa85dce785d8d33496
X-Request-ID: D9D2D62BB5EA4EBD8E54F1EF16F5925C-2147483648-2147516416
{
  "business_object": "ticket",
  "messages": [
    {
      "id": 2001,
      "contents": [
        {
          "field": "subject",
          "value": "bovine rhinotracheitis"
        }
      ]
    },
    {
      "id": 2002,
      "contents": [
        {
          "field": "subject",
          "value": "how much does pet insurance cost for cats?"
        }
      ]
    },
    {
```

```
      "id": 2003,
      "contents": [
        {
          "field": "subject",
          "value": "how much is a heater core replacement"
        }
      ]
    }
  ]
}
```

Example

Sample Response (no options specified):

```
{
    "results": [
        {
            "detected_language": "de",
            "id": 2001,
            "status": 91,
            "status_message": "no active model found for this language"
        },
        {
            "detected_language": "en",
            "id": 2002,
            "recommendation": [
                {
                    "score": 1,
                    "solutions": [
                        {
                            "field": "doc",
                            "value": "$15 and $18 per month"
                        },
                        {
                            "field": "article_id",
                            "value": "91"
                        },
                        {
                            "field": "data_source",
                            "value": "mindtouch"
                        }
                    ]
                },
                {
                    "score": 0.78,
                    "solutions": [
                        {
                            "field": "doc",
                            "value": "Acetylcholine (ACh)"
                        },
                        {
                            "field": "article_id",
                            "value": "85"
                        },
                        {
                            "field": "data_source",
                            "value": "mindtouch"
                        }
                    ]
                }
            ],
            "status": 0,
```

```
            "status_message": "ok"
        },
        {
            "detected_language": "en",
            "id": 2003,
            "recommendation": [
                {
                    "score": 1,
                    "solutions": [
                        {
                            "field": "doc",
                            "value": "$563 to $733"
                        },
                        {
                            "field": "article_id",
                            "value": "44"
                        },
                        {
                            "field": "data_source",
                            "value": "mindtouch"
                        }
                    ]
                },
                {
                    "score": 0.86,
                    "solutions": [
                        {
                            "field": "doc",
                            "value": "A boil, also called a furuncle, is a deep
folliculitis, infection of the hair follicle.It is most commonly caused by
infection by the bacterium Staphylococcus aureus."
                        },
                        {
                            "field": "article_id",
                            "value": "3"
                        },
                        {
                            "field": "data_source",
                            "value": "mindtouch"
                        }
                    ]
                }
            ],
            "status": 0,
            "status_message": "ok"
        }
    ]
}
```


Example

Sample Request with threshold option:

```
{
   "business_object": "ticket",
   "messages": [
   {
         "id": 2017,
         "contents": [
            {
               "field": "subject",
               "value": "How bad is the new book by J.K Rowling"
            }
```

```
        ]
      }
   ],
   "options":{
      "threshold":{
            "high_confidence_label": "high confidence custom label",
            "value":0.85,
            "low_confidence_label":"low confidence custom label"
       },
         "recommendation_top_k": 5
   }
}
```

Example

Sample Response with threshold option:

```
{
    "results": [
        {
            "status_message": "ok",
            "recommendation": [
                {
                    "confidence_label": "high confidence custom label",
                    "solutions": [
                        {
                            "value": 405314,
                            "field": "article_id"
                        },
                        {
                            "value": "jam",
                            "field": "datasource"
                        }
                    ],
                    "score": 0.9697179505973824
                },
                {
                    "confidence_label": "high confidence custom label",
                    "solutions": [
                        {
                            "value": 407327,
                            "field": "article_id"
                        },
                        {
                            "value": "jam",
                            "field": "datasource"
                        }
                    ],
                    "score": 0.9000195363090295
                },
                {
                    "confidence_label": "high confidence custom label",
                    "solutions": [
                        {
                            "value": 407845,
                            "field": "article_id"
                        },
                        {
                            "value": "jam",
                            "field": "datasource"
                        }
                    ],
                    "score": 0.8885381751826298
```

```
                },
                {
                    "confidence_label": "high confidence custom label",
                    "solutions": [
                        {
                            "value": 406863,
                            "field": "article_id"
                        },
                        {
                            "value": "mindtouch",
                            "field": "datasource"
                        }
                    ],
                    "score": 0.8533724315936498
                },
                {
                    "confidence_label": "high confidence custom label",
                    "solutions": [
                        {
                            "value": 407627,
                            "field": "article_id"
                        },
                        {
                            "value": "mindtouch",
                            "field": "datasource"
                        }
                    ],
                    "score": 0.8533724315936498
                }
            ],
            "status": 0,
            "detected_language": "en",
            "id": 2017
        }
    ]
}
```

9 Clustering API

This is the ML scenario type Clustering.

> → Tip
>
> See the blog post Get Insights on Historical Tickets with Service Ticket Intelligence.

Related Information

Cluster Messages [page 165]

9.1 Cluster Messages

Gain insights from unstructured data to identify trends or patterns in service tickets from the clusters and keywords you have created. It is also possible to upload a new dataset (in similar context) during inference to generate new clusters.

Key	Value
End-Point	Clustering
HTTP Method	POST
Path	text/cluster
Headers	Content-Type: application/json Authorization: Bearer <token> X-Request-ID: <unique string>

Related Information

Clustering API Input Attributes [page 166]
Clustering API Expected Response [page 167]
Clustering API Examples [page 169]

9.1.1 Clustering API Input Attributes

All input attributes are optional. By default, an empty request body payload will retrieve **all** clusters from the initial training data.

Attribute	Type	Description
`language`	String (optional)	Denotes the language of the model from which you fetch the clusters. The language is defined during file upload. This attribute is optional. If not specified, it defaults to English.
`mapping`	Mapping Object (optional)	Fields you use as input to generate new clusters. See Mapping Object [page 166].
`options`	Options Object (optional)	Allows you to perform additional operations during inference of clusters. See Options Object [page 167].

9.1.1.1 Mapping Object

A mapping object that describes which columns in the uploaded file should be used as inputs and outputs to create new clusters.

Attribute	Type	Description
`input`	Array<String>	A collection of strings referring to the column names in the uploaded file that are to be considered as input.
`output`	Array<String>	A collection of strings referring to the column names in the uploaded file that are to be considered as output.

9.1.1.2 Options Object

Attribute	Type	Description
`top_k_clusters`	Integer	Indicates the required number of clusters.
`cluster_language`	List	Indicates the language for clustering. Each language can have two or more clusters.
`cluster_groupby`	Dict	A dictionary of column name as the `key` and column value as the `value`. This attribute acts as a filter.

9.1.2 Clustering API Expected Response

As tokens are based on language, the response will cluster data based on languages identified from the dataset.

Attribute	Type	Description
`language`	Array of Language Clusters Objects	The results from the clustering of messages. Clusters are grouped by language. If dataset comprises of a mix of languages, for example en, de, and fr, there will be three groups. See Language Clusters Object [page 168].
`status`	Integer	Status code of the given ticket's classification, which would be **0** when successful.
`status_message`	String	User-friendly error message.

9.1.2.1 Language Clusters Object

Attribute	Type	Description
`clusters`	Array of Cluster Objects	Within each language cluster there will be **n** clusters, where **n** can refer to the total number of clusters or a specified number of `top_k_clusters` in the request. See Clusters Object [page 168].
`message`	String	If there are any errors during clustering this field will display the error reason.
`num_clusters`	Integer	Indicates the number of clusters. It has the same numbers as in `top_k_clusters` if that is set.
`obj`	String	Indicates ticket clusters.
`records_in_clusters`	Integer	Indicates the number of records in the cluster.
`total_tickets`	Integer	Indicates the total number of records in the dataset.

9.1.2.1.1 Clusters Object

Attribute	Type	Description
`conf`	Float	Indicates the confidence value of the cluster, for example, the level of coherence within that cluster.
`keywords`	List	Indicates the keywords along with the weights to indicate prominence.
`language`	String	Language of the text which forms the cluster.

Attribute	Type	Description
`tickets`	List	List of tickets that form the cluster. Each element in the list is a dictionary with key/value pairs. These pairs are the same that are provided as output during file upload. Alternatively, if no output was specified, these pairs will be the same as the columns specified in input fields.

9.1.3 Clustering API Examples

Example

Sample Code with No Inputs to Retrieve Clusters (retrieves all **clusters generated from the training data):**

Request Body

```
{}
```

Response Body

```
{
   "en":{
      "clusters":[
         "…"
      ],
      "message":"",
      "num_clusters":18,
      "obj":"ticket_clusters",
      "records_in_clusters":1243,
      "total_tickets":1243
   },
   "status":0,
   "status_message":"ok"
}
```

Example

Sample Code with Options Input (retrieves `top_k_clusters` **from the training data):**

Request Body

```
{
   "options":{
```

```
      "top_k_clusters":2,
      "cluster_groupby":{
         "column":"label",
         "value":[
            "biology"
         ]
      }
   }
}
```

Response Body

```
{
   "en":{
      "clusters":[
         "…"
      ],
      "message":"",
      "num_clusters":2,
      "obj":"ticket_clusters",
      "records_in_clusters":63,
      "total_tickets":104
   },
   "status":0,
   "status_message":"ok"
}
```

10 Common Status and Error Codes

Code	Message	Reason
0	`ok`	Success, all good.
1-39	-	Reserved.
40	`transformations during file upload is not supported`	Transformations not allowed during non-odata file upload.
41	`transformations only supported for classification type`	Transformations only supported for scenario type Classification and not for Recommendation.
42	`some of the output_category in transformations are missing from output in mapping`	Not all of the output_category fields in transformation section are present in the output of mapping section.
43	`odata info should not be empty`	OData object is not found in the request.
44	`odata credential should not be empty`	Credential object is not found in the request OData object.
45	`odata service url should not be empty`	OData service URL is not found in the credential object.
46	`odata authorization info should not be empty`	OData service authorization info is not found in the credential object.
47	`odata application info is should not be empty`	Application name is not found in the request.
48	`invalid application`	Application name is not valid or not approved to access OData endpoint.
49	`invalid odata template type`	Template is not a valid predefined template.
50	`invalid date format`	The date is not in acceptable format.
51	`invalid date / future date`	The date is invalid or future dated.

Code	Message	Reason
52	-	Reserved.
53	`either date_from or date_to is allowed`	Both *date_from* and *date_to* are not allowed in a single request.
54	`odata query options should not be empty or provide template`	Both OData query options and template are not found in request.
55	`odata entity name should not be empty`	Entity name is not found in OData query option object.
56	`odata select option should not be empty`	Select option is not found in OData query option object.
57	`input/output mapping is not same as select option`	Mapping input and output field names and OData query options select option are not matching each other.
58	-	Reserved.
59	`data extractor call failed`	There was an internal error calling data extractor.
60	`not enough unique sample data for training`	There are not enough unique samples of data that are required for training.
61	`not a classification model`	The specified model is not a classification model, features like disable or enable model category are only applicable for classification type model.
62	`invalid output field`	The field to be disabled or enabled is invalid, the specified field must match with the specified model.
63	`invalid category classes`	The category classes to be disabled or enabled are invalid, the specified category classes must exist in the specified model.
64	`nothing to modify in category classes, request already satisfied`	The category classes to be disabled are already disabled, or those to be enabled are already enabled, no further action is required.
65	`incompatible model version`	The specified model is deprecated and not eligible for disable or enable of model category class.
66	`number of enabled categories should be at least 2`	Disable model category class request is not allowed, because the number of enabled category classes is already at the minimum threshold.

Code	Message	Reason
67	`invalid threshold value`	The specified threshold for inference is invalid, the value of threshold should be between 0.0 to 1.0.
68-72	-	Reserved.
73	`failed to deactivate model`	The model to be deactivated is already INACTIVE.
74	`reference model version is incompatible`	The reference model version is not eligible for incremental training, the model algorithm is incompatible with current runtime version.
75	`confusion matrix not supported`	The specified model does not have or support confusion matrix details.
76	`exceeded tabulation max limit`	Maximum of three tabulation source-target is allowed per model.
77	`tabulation result not found`	Tabulation information is not found in model. Either tabulation is not requested during file upload, or the training is not yet started / completed.
78	`invalid tabulation source field`	Tabulation source input field should match with the Training tabulating input source field.
79	-	Reserved.
80	`exceeded models max limit`	For each subscription only a maximum of 20 training models is allowed.
81	`invalid scenario type`	The type of machine learning service to be trained against is not valid.
82	`invalid scenario business object`	The business object of the model to be trained for is not valid. Currently, the only possible business object is *ticket*.
83	`invalid scenario language`	The specified language of the training is not valid. The language is granted during customer on-boarding (provisioning), default is *EN* (English).
84	`reference model status is not valid for incremental upload`	The reference model status is not valid for incremental training, most likely the reference is INVALID.
85	`reference model status is not valid for incremental training`	The reference model status is not valid for incremental training, most likely the reference is INVALID, IN_TRAINING, or NEW. The reference model must be in model status READY / LOADING / ACTIVE for incremental training.
86-87	-	Reserved.

Code	Message	Reason
88	`invalid prediction field`	Text Classification and Recommendation inference field value should match with the ACTIVE model mapping input field.
89	`invalid upload file / bad format`	The uploaded training file is malformed.
90	`invalid model status`	The given model status is not valid.
91	`no active model found for this language`	The inference message content is in a language that does not have an active model. You should train or activate a model for that language.
92	`model not found`	The specified model ID (or reference model ID for incremental training) is not valid, or the model was corrupted.
93	-	Reserved.
94	`unable to persist model`	There was an internal error in which the model could not be stored persistently.
95	`unable to complete model training`	Model is not eligible for training. Model training must start from model status NEW.
96	`unable to activate model`	Model is not ready to be activated (still in model status NEW or IN_TRAINING) or already in model status ACTIVE.
97	`unable to delete model`	The model (or training file) is not in a removable state. Models with model status LOADING, ACTIVE or IN_TRAINING (less than 50 hours) cannot be deleted.
98	`bad request` or `bad request: <error details>`	The request is malformed. The specific error may be further provided in <error details> (`Failed to decode JSON object`), for example.
99	`unexpected error`	This is an unknown error.
101	`Simultaneous data extraction jobs are denied`	Data extraction already in progress, new data extraction job cannot be submitted.
90601	`Security error: Unable to authorize the request`	The request authorization header and/or access token is missing or invalid.

11 Glossary

Term	Definition
Business Object	A set of entities with common characteristics and common behavior that represent a specific view on well-defined and outlined business content (for example purchase order, sales order, and customer). Business objects are defined free of business functionality redundancies.
CF	Cloud Foundry.
CSV	Comma-separated values. A text file format which specifies tabular data.
KB	Knowledge Base. A repository of articles that provide knowledge on specific topics.
ML	Machine Learning.
UAA	User Account and Authentication.
URL	Uniform Resource Locator.
XS	SAP HANA Extended Application Services.

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